Quantifying the Impact of Temporal and Causal Structures on Narrative Similarity Retrieval

A Hybrid Event-Based Framework Combining Supervised Event Extraction and LLM-Inferred Temporo-Causal Relations for Structured Narrative Representation

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Buğra Sipahioğlu
14318334

Master Information Studies
data science
Faculty of Science
University of Amsterdam

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	UvA Supervisor
Title, Name	Alexandra Barancová
Affiliation	UvA Supervisor
Email	a.barancova@uva.nl



Abstract

Computational narrative analyses increasingly use retrieval models to identify similar stories, but separating narrative logic from surface-level entities remains challenging. Dense embeddings capture rich context yet suffer from entity bias, while explicit event-based structures improve explainability but achieve lower performance and omit narrative aspects such as temporal and causal links. It therefore remains unclear whether adding these relations can improve similarity retrieval. We address this with a hybrid framework that extracts events using a supervised transformer and infers relations via a large language model. These structural representations are embedded and evaluated on the Tell Me Again! dataset. Results show that while event-only models provide robust retrieval signals when named entities are removed, temporal and causal enrichments do not improve performance. Our findings quantify the limits of explicitly embedding temporo-causal structures and highlight the need for unified datasets with ground-truth annotations across narrative dimensions.

Keywords

narrative similarity, event extraction, temporal relations, causal relations, narrative embeddings, information retrieval

Github Repository

https://github.com/bgrsph/uva-thesis-computational-narrative-analysis

1 Introduction

Computational narrative analysis is increasingly used across domains such as economics, politics, psychology, medicine, law, and business [6, 8]. At the core, it builds on narratological frameworks such as Genette's triangle, where narrative comprises story (what happens), discourse (how it is told), and narrating (who is telling it) [26]. Computational approaches use natural language processing (NLP) to model these dimensions at scale, a task increasingly supported by Large Language Models (LLMs) because of their ability to connect linguistic features with theoretical narrative constructs [25]. This shift enables large-scale retrieval and QA systems to identify narrative similarity, the main task in this study: retrieving summaries of the same underlying story and recognizing structurally similar retellings despite surface-level differences [4, 11].

However, the domain faces two core challenges. The first is to distinguish surface-level lexical semantics from deep narrative structures. Standard models often over-rely on vocabulary or named entities (e.g., location names) rather than inherent plot logic [12]. As a result, models may fail to match alternative tellings of the same underlying story when they differ substantially in vocabulary, style, or named entities.

The second challenge is modeling the dynamic nature of the narratives. Here, identifying *what* happens (i.e., extracting events) can be insufficient without understanding *when* (temporality) and *why* (causality) they occur [21, 38]. Although state-of-the-art (SoTA) event-extractor models reach high performance on established benchmarks, extracting the temporal or causal relationships is challenging because it requires "common sense reasoning" to interpret these context-dependent relationships within narratives [21]. Then, manually annotating temporal and causal event relations from narratives becomes time-consuming [21, 38, 40].

Previous work has modeled narratives as event-character structures, LLM-extracted character roles, or dense summary embeddings [4, 8, 11, 25, 36]. However, these approaches expose a trade-off: dense embeddings preserve rich context but are less interpretable and can encode named entities, whereas explicit event-based representations are more interpretable but may lose context and omit temporal-causal structure [8, 10, 12]. Furthermore, [10] show that narrative-cloze embeddings, trained to predict masked events within event chains, perform poorly at identifying similar stories, which suggests that event chains alone may not capture how events are ordered and connected across a narrative. Although temporal-causal links offer one way to model such inter-event relations [21, 25], there is limited evidence on whether they improve event-based representations for narrative similarity retrieval. Moreover, extracting temporal-causal relations adds complexity and introduces two major vulnerabilities: upstream error propagation from misclassified events, and LLM hallucinations of spurious links [16]. Thus, their added value must be systematically tested against a robust event-only baseline.

To our knowledge, no study has systematically quantified the incremental contribution of temporal and causal modeling over event-only representations for narrative similarity. Addressing this gap allows us to assess whether structured, interpretable representations improve on event-only baselines and whether they remain competitive with high-capacity models that embed raw, unstructured text. We therefore combine supervised event extraction with LLM-based temporal and causal "commonsense" inference to evaluate their joint contribution. Then, the research (sub)question(s) are as follows: *To what extent does explicit narrative modeling through events and LLM-inferred temporal and causal relations improve the retrieval of narratively similar stories, compared with event-only, lexical, and implicit dense embedding baselines?*

- *SQ1:* Does event-based narrative representation outperform lexical baselines (BoW and TF-IDF) in narrative similarity retrieval?
- *SQ2:* What are the independent and joint effects of adding explicit temporal and causal relations to event-based representations?
- *SQ3:* To what extent do jointly modeled temporo-causal structures provide additional improvements over separately integrated temporal and causal augmentations?
- *SQ4:* Are the retrieval improvements observed when moving from event-only representations to temporally enriched, causally enriched, and jointly temporo-causal representations consistent across different embedding architectures (Qwen3-0.6B and E5-Mistral-7B-Instruct)?
- *SQ5:* How does the best-performing explicit structural model compare to a strong implicit dense narrative model (StoryEmb) and lexical baselines (BoW and TF-IDF)?

2 Related Work

2.1 Narrative Representation and Similarity

Narratives include multiple dimensions, such as agents, events, objects, locations, temporal frames, motivations or causal links [26]. Yet, none of these dimensions need explicit stating. Even when many remain implicit, a narrative still requires a minimal configuration in which "something happens to someone somewhere at some time for some reason and someone tells someone this story" [24]. This configuration provides a basis for computational narrative analysis by identifying recurring elements that can be detected, represented, and compared. Accordingly, prior work has represented narratives as event chains [3], plot structures [9], character roles [31], emotional arcs [28], and turning points [23]. While these approaches primarily model internal narrative mechanics, their structural abstractions provide a basis for comparing narratives through shared patterns, roles, events, or trajectories.

A computational approach to framing narrative similarity as an information retrieval task was introduced by [4], who represented stories through a set of event triggers and character roles. This abstraction improved retrieval over a Bag-of-Words baseline, but only modestly (8%). The authors attributed this to information loss and the absence of temporal or affective modeling. Later, [11] addressed this limitation through StoryEmb, an E5-Mistral model fine-tuned on the Tell Me Again! dataset [12] to retrieve alternative summaries of the same story. To reduce entity bias, they also trained on pseudonymized summaries in which names and locations were replaced. Although StoryEmb achieved strong retrieval performance (94.64 P@1), its drop under pseudonymized evaluation (82.60 P@1) suggests that dense narrative embeddings may still depend partly on surface-level cues rather than capturing narrative understanding.

More recently, [8] show that LLMs can extract structured narrative components and represent stories through role-based character embeddings. Although effective for clustering similar news articles, this approach remains character-centric and may overlook differences in temporal or causal progression. Together, these studies suggest that narrative similarity may not be fully captured by lexical overlap, dense summary embeddings, or role-based structure alone. What remains underexplored is whether explicitly modeling temporal and causal event structure adds retrieval value.

2.2 Event-based Narrative and Event Extraction

As narratives can be defined by the occurrence, order, and relationship of events, detecting events is a prerequisite both for representing narratives and, in turn, for detecting narrative similarity [11]. A common feature of SoTA event detection methods is that they utilize pretrained language models [16]. Although transformer-based models like BERT (variants) are common, their performance often differs depending on how events are theoretically defined and operationalized within specific ontologies [16]. In addition, transformer-based semantic event extraction has been shown to outperform LLM-based prompting methods in structured event classification, particularly when the ontologies are larger and more fine-grained [14].

Regarding the training resources of these models, large-scale benchmark datasets such as Automatic Content Extraction (ACE) and a MAssive eVENt detection dataset (MAVEN) provide fine-grained event ontologies that are beyond simple event triggers (verbs), including abstract event types (e.g., hurricane, war) across diverse domains [35, 38]. Unlike ACE, MAVEN offers a publicly available alternative without licensing restrictions. In particular, MAVEN enables supervised transformer models to learn mappings between lexical cues and structured event classes across a broad general-domain ontology of 168 types (e.g., Attack, Transport) [38].

2.3 Temporal and Causal Structure in Narratives

While event detection has reached a relatively stable level of performance, the annotation of semantic links between events remains substantially more challenging. Event relation extraction (ERE) models the temporal, causal, and coreference links between events, which are central to applications such as story generation, commonsense reasoning, question answering, and narrative forecasting [18, 37]. For narrative analysis, these links are especially important because coherence depends not only on which events occur, but on how they are temporally ordered and causally connected [21].

2.3.1 Modeling Temporal Relations. Temporal relation extraction is commonly studied via the TimeML framework, which operationalizes temporal expressions through TLINK annotations, based on Allen's interval algebra [27]. Subsequent benchmarks, such as TempEval, further standardize temporal relation modeling as a pairwise classification problem over annotated events [34]. More recent annotation efforts, like THYME, simplify the temporal link inventory to a smaller set of core relations (e.g. *Before*, *Overlap*) [32]. These frameworks show that temporal structure is essential for event ordering, yet local or sentence-bounded annotations can miss longer-range narrative event progressions [22].

2.3.2 Modeling Causal Relations. Unlike temporal relations, causal relation modeling remains less standardized. [20] note that while temporal modeling has achieved consensus through TimeML, annotation of causal relations lacks a unified framework. They introduce CLINK annotations inspired by TimeML to capture directed causal relations between events into categories like *Cause*, *Enable*, and *Prevent*. However, causality often depends on common sense reasoning and contextual inference rather than purely lexical cues [21]. In addition, causal relations impose temporal constraints: a cause must temporally precede or overlap with its effect [21].

2.3.3 Joint Modeling of Temporal and Causal Relations. Although temporal and causal relations are often studied independently, they are inherently intertwined in narrative structure [20, 21]. CaTeRS addresses this overlap by proposing a unified temporo-causal annotation scheme, where event-event links encode both explanatory relations and temporal ordering [21]. Its main assumption is that causal relations carry temporal implications: if one event causes or enables another, it must precede or overlap with its effect. This makes CaTeRS relevant for modeling narrative coherence through both why events occur and how they unfold over time. For instance, *Cause* and *Before* can be unified as *Cause_Before*.

2.3.4 LLMs as Relation Extractors. Despite the annotation frameworks, capturing semantic links between events remains challenging: temporal and causal annotation between events requires greater

interpretive effort and yields only moderate annotator agreement [21]. Moreover, while encoder-only models such as BERT are competitive for localized extraction, they are less suited to long-range, cross-sentence dependencies [6]. Since temporal and causal relation modeling requires discourse-level reasoning across the text, these constraints motivate our use of generative LLM-based inference.

2.4 Embedding Narratives

Transforming narratives into vector spaces often involves separating the fabula (underlying events and logic) from the discourse [11, 17]. Since this distinction is challenging to operationalize and narratives are not standardized, researchers adapt a range of NLP encoding strategies. These include sparse lexical representations for surface-level event or term overlap [4], dense embeddings for broader semantic similarity [39], fine-tuned dense encoders for implicit narrative similarity [11], and explicitly structured representations based on extracted narrative elements [8, 10, 15].

2.4.1 Lexical and Early Embedding Baselines. Before widespread adoption of contextual encoders, semantic similarity was commonly modeled using lexical vector space representations such as Bag-of-Words (BoW) [4] and TF-IDF [13]. These approaches represent texts as sparse vectors based on word frequency statistics and measure similarity through cosine distance. While computationally efficient and interpretable, they are heavily dependent on surface-level lexical overlap and therefore sensitive to named entities and stylistic variation [5]. Early dense approaches such as Doc2Vec generate fixed-length document vectors, but unlike transformer encoders they do not dynamically model how each token's meaning depends on surrounding tokens [11].

2.4.2 LLMs as Encoders. Recently, instruction-tuned embedding models such as E5 have become foundational for semantic retrieval because they are explicitly trained to align queries and relevant texts in embedding space [36]. This retrieval objective makes E5 especially relevant for narrative similarity, where the task is to retrieve alternative summaries of the same underlying story. Accordingly, recent narrative-representation studies use E5-based models: [11] build StoryEmb on E5-Mistral for same-story retrieval, while [8] use E5-large to compare structured narrative representations in news. These studies also report E5-based models outperforming alternatives such as Sentence-BERT-style encoders, Doc2Vec, MiniLM, and BERTopic in their respective narrative retrieval or clustering settings [8, 11]. More recent embedding models extend this retrieval-oriented direction beyond E5. Qwen3-Embedding offers instruction-aware embeddings with long context windows and comparable benchmark performance [41]. Although not narrative-specific, its inclusion helps test robustness by assessing whether structural narrative representations remain effective across modern embedding architectures.

3 Methodology

We test whether explicit narrative structure improves retrieval of stories with similar underlying narratives over raw text alone. We use two datasets: MAVEN [38] provides the event ontology and supervised training resource for event extraction, while TMA [12] provides the story-summary corpus, where summaries of the same work are treated as relevant matches. The pipeline includes event extraction, temporal and causal relation inference, representation construction, embedding, retrieval, and evaluation, as shown in Figure 1.

3.1 Datasets

3.1.1 MAVEN: Event Extraction Dataset. MAVEN provides a large-scale, fine-grained event ontology with 168 event types across diverse domains, where event triggers are annotated and mapped to abstract event categories [38]. This is useful for the present study because lexically different triggers that express similar narrative events, such as "attack," "assault," or "bombing," can be mapped to a shared event type, such as *Attack*. In this study, events are extracted using a BERT+CRF model trained on the MAVEN training split, which has 2,913 documents. Following MAVEN, an *event* is a specific occurrence involving participant(s), an *event trigger* is the word (or phrase) that expresses it, and an *event type* is its semantic category. MAVEN was used unchanged from its official release, as it does not require any additional cleaning.

3.1.2 Tell Me Again!: Narrative Similarity Corpus. The Tell Me Again! (TMA) dataset is a narrative similarity corpus containing 96,831 summaries from 29,505 distinct stories, primarily movies as well as books and other narrative works [12]. Because each work can have multiple independently written or translated summaries, TMA is suited to retrieval-based narrative similarity evaluation. The summaries were harvested from five Wikipedia language editions: English, German, French, Italian, and Spanish, with non-English summaries machine-translated into English. TMA also provides a pseudonymized variant in which named entities are replaced with alternatives like *Entity X* or *Location Y*.

3.1.3 Domain Alignment of the Datasets. Although MAVEN provides a broad general-domain event ontology, it is not perfectly aligned with TMA. MAVEN contains many high-impact real-world events, such as military conflict, hurricanes, and civilian attacks [38], whereas TMA contains summaries of films, books, and biographies [12]. To assess this mismatch, we compared the MAVEN gold event-type distribution with the event types detected when the MAVEN-trained extractor was applied to TMA. As shown in Figure 2, the extractor identifies 154 of MAVEN's 168 event types in TMA, while the unused types account for only approximately 0.33% of MAVEN event mentions. However, Figure 3 shows that the prevalent event types differ: MAVEN is led by *Causation*, *Process_start*, and *Attack*, whereas Tell Me Again! is led by *Know*, *Motion*, and *Arriving*. Overall, while MAVEN's event distribution does not match the extracted event distribution of TMA, it offers sufficient event ontology coverage for the present study.

3.2 Experimental Setup

In essence, our pipeline tests whether structured event representations improve narrative similarity retrieval. First, summaries from TMA dataset are converted into event tuples containing an event trigger and event type using a supervised BERT-based extractor trained on MAVEN dataset [38]. Second, a Llama-3-8B-Instruct model [1] infers temporal, causal, and joint temporo-causal relations between the extracted events under zero-shot prompting.

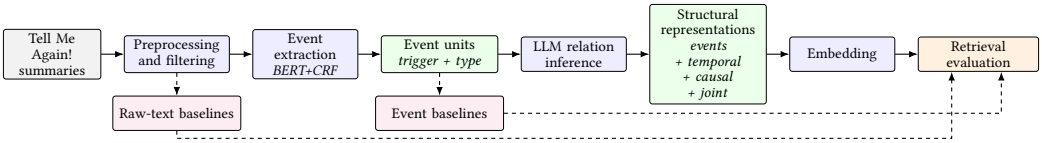


Figure 1: Overview of the pipeline for constructing, embedding, and evaluating event-based narrative representations. Colors indicate data inputs in gray, processing steps in blue, structural outputs in green, baselines in purple, and evaluation in orange.

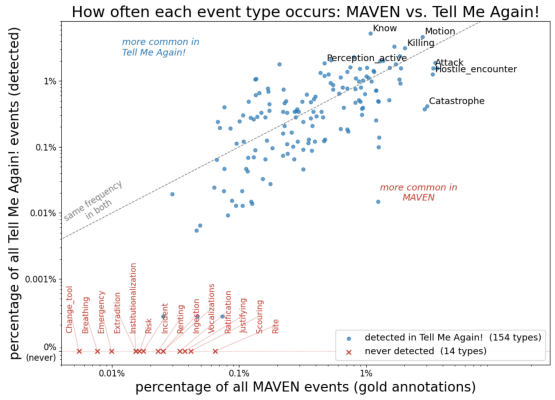


Figure 2: Ontology coverage comparison between MAVEN gold annotations and event types detected in non-pseudonymized TMA test split. Each point represents one MAVEN event type.

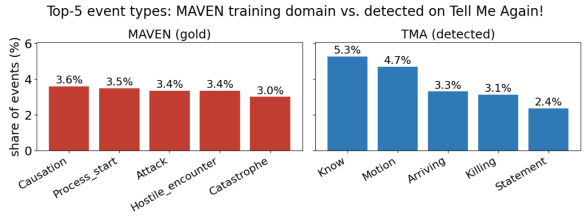


Figure 3: Top-five event-type distributions in MAVEN gold annotations and in event types detected on full non-pseudonymized TMA test split.

Third, the resulting structural representations are linearized, then embedded with E5-Mistral-7B-Instruct [36] and Qwen3-0.6B [41], and evaluated through cosine-similarity retrieval on TMA, where summaries of the same work are treated as relevant matches.

3.2.1 Data Preparation. We use the official TMA test split, which contains 10,914 summaries from 2,951 works, rather than the train, development, or full corpus. This avoids data leakage in the comparison with StoryEmb, our main narrative-specific dense embedding baseline, because StoryEmb was trained on the TMA training data [11]. Test split summaries contain 17.82 sentences and 366 words on average (see Appendix A). Following the preprocessing used by [12] for TMA, near-duplicate summaries across languages were removed using BERTScore ($\geq$ 0.6) [12]. This step reduces cases where different Wikipedia language versions are effectively direct translations rather than independent retellings. Cluster sizes also become slightly smaller after deduplication, decreasing from a mean of 3.70 summaries per story to 3.31 (see Appendix A). Finally, works with fewer than two usable English summary representations were excluded, since they cannot form relevance clusters for retrieval evaluation. After this, the initial experimental corpus contains 9,385 summaries from 2,804 works.

3.2.2 Event Extraction.

Training the Event Extractor. Event triggers and event types are extracted using the BERT+CRF model trained on MAVEN [38]. This architecture is selected because it is the strongest reported MAVEN baseline that outperforms alternative supervised configurations in trigger classification. The model is fine-tuned on the MAVEN training split for five epochs using AdamW, a learning rate of 5×10^{-5}, a maximum sequence length of 128 subword tokens, a batch size of 128, and random seed of 0. The resulting checkpoint achieves 67.3 F1 on the MAVEN trigger-classification validation set (not used during its training), which closely reproduces the 67.8 F1 reported by [38]. After fine-tuning, this checkpoint is frozen and reused unchanged for all event extraction on TMA.

Extracting Events from TMA. Because MAVEN is formulated as a sentence-level event detection task, each TMA summary is processed sentence by sentence before inference. Sentence boundaries are taken from the official TMA structure for the non-pseudonymized corpus and produced with a rule-based terminal-punctuation splitter for the pseudonymized corpus, since the boundaries were unavailable. The MAVEN-trained BERT+CRF detector is then applied to each sentence independently in batches of 32. This fine-tuned transformer-based setup is used because supervised models better enforce predefined event ontologies, while prompt-based LLMs are more prone to type misalignment and invalid event classes [14]. After the aforementioned corpus-level filters, the longest retained sentence contains 126 BERT subword tokens, which is below the model's 128-token input limit. Exemplar extraction outputs and actual outputs are provided in Appendices B and D, respectively.

Post-Extraction Filtering. After event extraction, summaries with fewer than five detected events are removed, since very sparse event sequences provide limited material for structural representation. Because no established minimum event-count threshold exists for narrative similarity retrieval, the choice is motivated by adjacent narrative-event work, where [21] show that five-sentence stories can support temporal-causal annotation (CaTeRS), which averages to 8.5 events per story. An exploratory analysis showed that the five-event threshold retains 89.7% of summaries, compared with 77.1% under a nine-event threshold (see Appendix C). Then, the five-event minimum is used to preserve brief summaries while excluding cases with too little event structure. The two-summary-per-work

requirement is then re-applied, resulting in 8,222 summaries from 2,589 works. All filtering done to mitigate data attrition can be found in Appendix K.1.

3.2.3 *Temporal & Causal Relation Extraction.*

Overview. Following event extraction, we annotate inter-event temporal and causal relations. Because these relations rely on commonsense and discourse-level narrative understanding rather than explicit lexical cues [21], and also since TMA lacks gold temporo-causal annotations for supervised training, we use an LLM for zero-shot relation inference. This allows temporal, causal, and joint temporo-causal links to be inferred from broader narrative context without fine-tuning a domain-specific relation extractor [16, 25].

LLM Details. We use Llama-3-8B-Instruct to annotate temporal and causal relations between extracted events [1]. This open-source model achieves zero-shot performance comparable to proprietary systems such as GPT-3.5 while maintaining narrative understanding and computational efficiency [8]. Moreover, its open weights enable reproducibility. Recent work further shows that Llama-3-8B can reliably operationalize narratological constructs into structured outputs, achieving performance comparable to GPT-4 in narrative analysis tasks [25]. To mitigate the inherent stochasticity of LLMs and ensure stability, we run model inference without sampling. By forcing the model to select the highest probability token at each step, the extraction process becomes reproducible and deterministic [8]. To ensure determinism, we disable sampling (*do_sample=False*), so the model performs greedy decoding where it selects the highest-probability token at each step. With the sampling off, temperature, top-p, and top-k are not applied, and we use a single beam (*num_beams=1*), so no beam search is performed.

Prompt Details. To control the output format and reduce hallucinated or inconsistently formatted relations, the LLM is prompted to return predictions strictly as a structured JSON object, a constraint shown to be effective for extracting narrative roles [8]. Each prompt is zero-shot and consists of a system message specifying the task-specific relation codebook and JSON schema, followed by a user message containing the summary with extracted events inlined as `[eID|trigger|EVENT_TYPE]` markers. Following the constraints from [2], prompts explicitly forbid conversational filler, explanations, invented event IDs, and out-of-schema labels. This is particularly important because inter-event relation extraction requires composite objects linking a source event ID, target event ID, and relation label, and LLM formatting reliability is known to decrease for such structured outputs [30]. Zero-shot prompting is used because TMA lacks gold temporal, causal, and temporo-causal relation annotations from which balanced in-domain few-shot examples could be sampled; this avoids the predictive bias that can arise from unrepresentative few-shot exemplars [33]. Our prompts also begin with a task-specific role assignment, supported by the evidence that shows task-aligned expert personas can produce positive or (at least) non-significant performance changes relative to no-persona baselines, particularly for Llama models [7].

All prompts are serialized with the Llama-3 chat template, using the appropriate headers and end-of-turn markers to separate system instructions from input content and to place the model at the intended JSON-generation point [1]. All relational conditions are prompted independently, each with its own label codebook. Label names are derived from [21], and definitions are operationalized. During inference, generation is limited to the remaining Llama-3 context, `max_new_tokens = 8192 - |prompt token length|`, and terminated using the standard Llama-3 end-of-turn tokens. Outputs are parsed as JSON and validated against the condition-specific label set and event-ID inventory. Malformed, out-of-codebook, or hallucinated relations are discarded. The prompt templates with the codebooks for relations are provided in Appendix E, exemplar prompts with annotated events can be found in Appendix F.

Annotating the Relations via LLM. Given a summary with a fixed inventory of extracted events and a condition-specific codebook of relation labels, the LLM is asked to assign semantic links between already extracted events. Three prompted relation conditions are generated independently: temporal, causal, and joint temporo-causal. The label codebook of each relation type can be found within the prompt templates, in Appendix E. Note that the joint temporo-causal condition uses a fused CaTeRS-inspired label set in which causal relations encode their temporal implication, such as *Cause_Before* or *Enable_Overlaps*, while also allowing temporal fallback labels when no causal relation is inferred [21]. In addition, a fourth independent temporal+causal representation is derived deterministically by combining the separately generated temporal and causal outputs, without making an additional LLM call. This design separates two questions: whether temporal and causal structures contribute independently, and whether a joint temporo-causal ontology (CaTeRS) provides additional signal beyond their post-hoc combination. Exemplar prompt and JSON outputs are provided in Appendix F and Appendix D, respectively.

Post-Annotation Filtering. After relation inference, summaries are removed only when annotation fails in at least one prompted condition, due to JSON parse error or context-window overflow by the LLM. The two-summary-per-work requirement is then re-applied to remove resulting singletons, resulting in 6,189 summaries from 2,180 works. Hallucinated, invalid, or out-of-codebook relation triples are removed from summaries, but the summaries themselves are kept. Details of the data processing to prevent data attrition can be found in Appendix K.1.

3.2.4 *Embedding and Retrieval.*

After event extraction and relation inference, each summary is converted into one or more linearized textual representations. Each representation is constructed as a canonical sequence containing the extracted event units in textual order, optionally followed by temporal, causal, independent temporal+causal, or joint temporo-causal relation triples depending on the experimental condition. We linearize relation triples in a hybrid symbolic-textual form: event IDs preserve relation references when the same event participates in multiple relations, and attached triggers and event types provide the natural-language semantics needed by dense text embedders for retrieval.

We encode the linearized representations using three instruction-aware sentence encoders. First, E5-Mistral-7B-Instruct is applied to all structural representation conditions [36]. For cross-architecture robustness, we also use Qwen3-Embedding-0.6B, a smaller decoder-only, long-context embedding model [41]. The 0.6B variant is selected to test whether structural effects persist beyond E5-Mistral

while remaining computationally feasible for repeated embedding processes across the full ablation grid. Finally, StoryEmb, an E5-Mistral model fine-tuned on pseudonymized TMA narratives, is used for raw-text retrieval as a narrative-specific dense baseline [11], which provides a ceiling estimation for narrative retrieval.

All encoders receive the same StoryEmb-style retrieval instruction: "Retrieve stories with a similar narrative to the given story" [11]. Examples of linearized structural representations are provided in Appendix G. Inputs longer than 4,096 tokens are excluded before embedding, since E5-Mistral performance deteriorates beyond its recommended 4,096-token window [36]. Encoder outputs are L2-normalized, and retrieval is performed by computing pairwise cosine similarity for each representation condition, masking self-matches, and ranking candidates by descending similarity.

3.3 Evaluation

3.3.1 Evaluation Framework and Metrics. We evaluate structural enrichment in a retrieval framework, defining narrative similarity as retrieving alternative summaries of the same underlying story from a shared candidate corpus [4, 11]. For each query, all other TMA test summaries are ranked by cosine similarity after masking self-matches. Since the non-pseudonymized and pseudonymized corpora produce different surviving summaries after preprocessing —applied to each of them separately— all results are computed on their matched intersection: 4,315 summaries from 1,658 stories, where each retains at least one same-work relevant summary in both settings (see Appendix K.1).

We utilize top-result and ranking-quality metrics to capture different aspects of performance. Precision@1 (P@1) is used to assess strict accuracy, measuring whether the model's top choice is a correct alternative telling, which mirrors realistic search expectations [4, 11]. We also report Hits@10 as a more forgiving top-k metric, which measures whether at least one correct alternative appears among the ten highest-ranked candidates, in line with prior structural-semantic embedding evaluations [29]. To ensure comparability with StoryEmb results [11], we report the same retrieval metrics used in their prior evaluation: R-Precision, Mean Average Precision (MAP), and Normalized Discounted Cumulative Gain (NDCG) [11]. These metrics are well suited to the Tell Me Again! setup because each query may have a different number of relevant summaries. R-Precision accounts for this by evaluating precision at R, where R is the number of gold-relevant summaries for the query. MAP and NDCG assess whether all relevant summaries of the same work are ranked highly [11, 19].

3.3.2 Ablation Conditions. We compare six representations within the same retrieval framework: raw text, event-only, event+temporal, event+causal, event+temporal+causal, and event+temporo-causal. Raw text serves as the unstructured baseline, while the other conditions are derived from the event extraction and relation inference pipeline. Each condition is evaluated with both encoders to test whether structural effects depend on the embedding architecture.

3.3.3 Lexical Baselines. We include lexical baselines to measure how much retrieval performance can be explained by surface overlap alone. Following [4], the textual summaries are first preprocessed through stopword removal and lemmatization before being converted into sparse Bag-of-Words (BoW) and TF-IDF representations. Full-text Bag-of-Words (BoW) [4] and TF-IDF [13] represent each summary using its original text where they capture named entities, repeated vocabulary, and other surface-level cues. In addition, we include event-trigger BoW and TF-IDF baselines, where each summary is represented only by the triggers extracted by the event detection model. These event-level diagnostic baselines isolate whether the event extraction step preserves retrieval-relevant story content before temporal or causal relations are added.

3.3.4 Dense Embedding Baselines. We compare against dense baselines that encode the original summaries directly and therefore retain named entities, stylistic cues, and global semantic context. E5-Mistral [36] and Qwen3-0.6B [41] serve as raw-text dense baselines for assessing how much retrieval performance is achieved without explicit event or relation modeling. We also compare against StoryEmb, E5-Mistral model fine-tuned on the TMA for same-story retrieval [11]. Since StoryEmb is both raw-text based and trained specifically on TMA, it serves as a narrative-specific ceiling rather than a directly equivalent baseline. Thus, we evaluate whether explicit event and relation representations provide interpretable retrieval signal and whether our enrichment narrows the gap to task-specific dense retrieval.

3.3.5 Manual Evaluation of Event Extraction. Since TMA lacks gold event annotations, we manually evaluate the BERT+CRF extractor on 30 randomly sampled summaries containing 5-10 sentences, on non-pseudonymized version of test split of TMA. Manual annotations are compared with model predictions using micro-averaged precision, recall, and F1 under exact span matching. We evaluate trigger identification, where only the span must match, and trigger classification, where the MAVEN event type must also match, to assess whether retrieval errors may partly reflect event-extraction noise.

4 Results

4.1 SQ1: Event-based representations versus lexical baselines

We show that under pseudonymization, event-only representations—independent of embedder selection—outperform full-text lexical baselines across all metrics. This indicates that event-only representations are more robust than full-text BoW and TF-IDF once named-entity cues are removed. As Table 1 shows, on Hits@10, the best event-only model reaches 0.3191, compared with 0.1900 for full-text TF-IDF and 0.1638 for full-text BoW. Yet, event-trigger TF-IDF is stronger than the best event-only model, at 0.3815 Hits@10.

4.2 SQ2: Independent effects of temporal and causal relations

As Table 2 summarizes, relation enrichment does not improve retrieval over event-only representations. Qwen3-0.6B gains negligibly for temporal+causal relations on all non-pseudonymized metrics except P@1, while all relation-enriched variants decrease performance under pseudonymization. For E5-Mistral, all relation-enriched conditions degrade performance in both settings. Although relation enrichment sometimes fixes incorrect event-only top-1

Table 1: SQ1, SQ4, SQ5 - Overall retrieval performance across baseline and structural representation conditions on the TMA.

		Non-Pseudonymized					Pseudonymized				
Representation	Vectorizer/Encoder	P@1	Hits@10	R-Prec.	MAP	NDCG	P@1	Hits@10	R-Prec.	MAP	NDCG
BoW (full text)	Count vectorizer	0.9154	0.9740	0.8637	0.8990	0.9312	0.0732	0.1638	0.0579	0.0801	0.2273
TF-IDF (full text)	TF-IDF vectorizer	0.9374	0.9859	0.8958	0.9265	0.9512	0.0811	0.1900	0.0668	0.0996	0.2595
BoW (event triggers)	Count vectorizer	0.2058	0.3235	0.1503	0.1748	0.3138	0.1919	0.3015	0.1390	0.1619	0.2997
TF-IDF (event triggers)	TF-IDF vectorizer	0.2440	0.4097	0.1844	0.2147	0.3568	0.2250	0.3815	0.1674	0.1973	0.3387
Full text	Qwen3-0.6B	0.9548	0.9852	0.9226	0.9438	0.9622	0.7108	0.8658	0.6197	0.6790	0.7671
Full text	E5-Mistral	0.7249	0.8345	0.5985	0.6420	0.7336	0.3502	0.5258	0.2669	0.3024	0.4401
Full text	StoryEmb	**0.9854**	**0.9975**	**0.9707**	**0.9806**	**0.9875**	**0.7736**	**0.9031**	**0.6883**	**0.7397**	**0.8138**
Event-only	Qwen3-0.6B	0.2162	0.3479	0.1510	0.1778	0.3195	0.1963	0.3191	0.1421	0.1644	0.3044
Event + temporal	Qwen3-0.6B	0.2058	0.3462	0.1484	0.1738	0.3165	0.1877	0.3117	0.1341	0.1580	0.2992
Event + causal	Qwen3-0.6B	0.1984	0.3319	0.1434	0.1676	0.3091	0.1824	0.3013	0.1326	0.1556	0.2946
Event + temporal + causal	Qwen3-0.6B	0.2072	0.3543	0.1517	0.1780	0.3207	0.1905	0.3145	0.1371	0.1615	0.3022
Event + joint temporo-causal	Qwen3-0.6B	0.2035	0.3409	0.1464	0.1722	0.3145	0.1847	0.3087	0.1308	0.1553	0.2967
Event-only	E5-Mistral	0.1638	0.2804	0.1177	0.1379	0.2775	0.1587	0.2677	0.1139	0.1351	0.2730
Event + temporal	E5-Mistral	0.0718	0.1365	0.0534	0.0639	0.1906	0.0707	0.1328	0.0485	0.0586	0.1861
Event + causal	E5-Mistral	0.0681	0.1249	0.0475	0.0579	0.1834	0.0630	0.1228	0.0449	0.0552	0.1814
Event + temporal + causal	E5-Mistral	0.0366	0.0709	0.0270	0.0335	0.1535	0.0276	0.0656	0.0211	0.0278	0.1484
Event + joint temporo-causal	E5-Mistral	0.0744	0.1499	0.0559	0.0666	0.1941	0.0681	0.1289	0.0515	0.0597	0.1864

Note. Results are computed on the matched intersection of the non-pseudonymized and pseudonymized evaluation sets. This matched set contains 4,315 summaries from 1,658 stories, with each query having at least one gold-relevant neighbor in both conditions. Bold values indicate the best score within each metric and pseudonymization condition. Underlined values indicate the best explicit structural representation within each metric and pseudonymization condition.

matches, it more often breaks cases where event-only retrieval was already correct (see Table 12 in Appendix J.4).

4.3 SQ3: Temporo-Causal vs Temporal+Causal

We find that jointly modeled temporo-causal relations do not provide a reliable advantage over independently combined temporal+causal relations. Table 2 shows that Qwen3-0.6B performs worse with joint temporo-causal relations than with temporal+causal relations across all metrics in both (non-)pseudonymized settings. For E5-Mistral, the joint condition outperforms temporal+causal across all metrics, but both remain below the event-only baseline.

4.4 SQ4: Robustness across embedding architectures

We see that the structural retrieval results are encoder-dependent, but relation enrichment remains ineffective across architectures. As Table 1 shows, Qwen3-0.6B consistently outperforms E5-Mistral across all event-based structural representations, with and without pseudonymization. However, Qwen3-0.6B shows only negligible and inconsistent benefits for temporal+causal relations in the non-pseudonymized setting, while all relation-enriched structures degrade performance under pseudonymization. For E5-Mistral, every relation-enriched condition performs worse than event-only in both settings.

4.5 SQ5: Best explicit structural model versus StoryEmb and full-text baselines

As shown in Table 1, we find that explicit structural models provide measurable retrieval signal but remain far below raw-text dense baselines. The best structural condition (under most metrics) is Qwen3-0.6B temporal+causal in the non-pseudonymized setting and Qwen3-0.6B event-only under pseudonymization.

4.6 Manual Evaluation of Event Extraction

Table 3 reports a manual evaluation of the BERT+CRF extractor on 30 randomly sampled *Tell Me Again!* summaries containing 5-10 sentences. The extractor identifies event triggers (therefore the spans) comparably well, but performance drops when the MAVEN event type must also match. This suggests that downstream errors may partly reflect noise from event type assignment.

4.7 Error Analysis

4.7.1 Structural Confusion in Top-1 Retrieval Errors. To test whether retrieval errors are random or carry structural signal, we analyze incorrect top-1 matches for every encoder-condition pair. For each failed query, the wrongly retrieved summary is compared with a random non-matching summary using four measures: event-type inventory overlap, event-type order overlap, trigger inventory overlap, and trigger order overlap. Across all encoder-condition pairs, wrong matches score significantly higher than random matches across all four measures ($p<0.001$; Appendix H.1). This indicates that wrong retrievals are structurally systematic, rather than random: they often share a higher-level event skeleton with the query, either through overlapping event inventories or similar event order. See Appendix H.2 for exemplars.

4.7.2 Reduced Discriminability after Relation Enrichment. To examine whether relation enrichment introduces noise, we test how adding temporal and causal links affects the separability of summary representations. We see that relation-enriched Qwen3-0.6B embeddings have higher average pairwise similarity and lower standard deviation than the event-only condition, which means that summaries become more similar to one another in the embedding space. Moreover, the inferred relation labels are also highly concentrated where a single label accounts for most links in each relation setting, with 84.8% of temporal links labeled *Before*, 91.1% of causal links labeled *Cause*, and 72.9% of joint temporo-causal links labeled *Cause_Before*. Most of these links also connect adjacent events. Hence, temporal and causal relations reduce the discriminatory

Table 2: SQ2, SQ3 - Incremental retrieval change from adding relation structure to event-only representations on the TMA.

		Non-Pseudonymized					Pseudonymized				
Encoder	Added structure	ΔP@1	ΔHits@10	ΔR-Prec.	ΔMAP	ΔNDCG	ΔP@1	ΔHits@10	ΔR-Prec.	ΔMAP	ΔNDCG
Qwen3-0.6B	Temporal	-0.0104	-0.0016	-0.0026	-0.0040	-0.0031	-0.0086	-0.0074	-0.0080	-0.0064	-0.0053
Qwen3-0.6B	Causal	-0.0178	-0.0160	-0.0076	-0.0102	-0.0104	-0.0139	-0.0178	-0.0094	-0.0088	-0.0098
Qwen3-0.6B	Temporal + causal	**-0.0090**	**+0.0065**	**+0.0007**	**+0.0002**	**+0.0012**	**-0.0058**	**-0.0046**	**-0.0049**	**-0.0029**	**-0.0022**
Qwen3-0.6B	Joint temporo-causal	-0.0127	-0.0070	-0.0046	-0.0056	-0.0051	-0.0116	-0.0104	-0.0112	-0.0091	-0.0077
E5-Mistral	Temporal	-0.0920	-0.1439	-0.0643	-0.0740	-0.0869	**-0.0881**	**-0.1349**	-0.0654	-0.0765	-0.0869
E5-Mistral	Causal	-0.0957	-0.1555	-0.0702	-0.0800	-0.0942	-0.0957	-0.1448	-0.0690	-0.0799	-0.0915
E5-Mistral	Temporal + causal	-0.1272	-0.2095	-0.0907	-0.1044	-0.1241	-0.1312	-0.2021	-0.0928	-0.1073	-0.1246
E5-Mistral	Joint temporo-causal	**-0.0895**	**-0.1305**	**-0.0617**	**-0.0712**	**-0.0835**	-0.0906	-0.1388	**-0.0624**	**-0.0754**	**-0.0865**

Note. Values are deltas from the event-only representation under the same encoder. Positive values indicate improvement; negative values indicate degradation. Results use the matched non-pseudonymized and pseudonymized evaluation set of 4,315 summaries from 1,658 stories. Bold marks the largest delta within each encoder block and metric.

Table 3: Manual evaluation of BERT+CRF event extraction on the TMA summaries vs reported MAVEN evaluation.

Data	Task	P	R	F1	Num of Gold	Num of Pred.
TMA	Identification	73.2	71.0	72.1	431	418
TMA	Classification	53.8	52.2	53.0	431	418
MAVEN	Classification	65.0	70.9	67.8	21,835	–

Note. Values are micro-averaged with exact trigger-span matching. Identification requires sentence and span match; classification additionally requires the event type match. The MAVEN row has the reported results by [38], prediction counts and identification scores are not reported.

qualities of summaries. Analysis on embedding-space similarity, relation-label distributions, adjacency patterns, and query-level retrieval transitions are reported in Appendix J.

5 Discussion

We show that event-only representations are robust when named entities are removed, while LLM-inferred temporal and causal relations generally reduce retrieval performance. We discuss this as a trade-off between robustness to entity bias and the limited discriminative value of relation enrichment.

5.1 The Tradeoff Between Robustness and Entity Bias

The pseudonymized evaluation qualifies the large performance gap observed in the main results. Although explicit structural representations perform below full-text baselines when entity cues are available, this comparison partly reflects the advantage of named-entity overlap in raw-text retrieval. Once character names, locations, and other named entities are replaced, full-text lexical performance degrades: raw-text TF-IDF falls to 0.0811 P@1, and BoW to 0.0732 P@1. By contrast, the event-based representations are anonymized by design, since summaries are represented only through event triggers, event types, and event-event relations. This makes them more robust under pseudonymization: event-trigger TF-IDF reaches 0.2250 P@1, while the Qwen3-0.6B event-only representation reaches 0.1963 P@1. Thus, although explicit structural models do not compete with dense full-text models overall, they outperform full-text lexical baselines once entity cues are removed.

This pattern helps explain why the present findings differ from earlier structured approaches to narrative retrieval. [4] reported that adding narrative structure improved P@1 from 0.558 to 0.637 on movie remakes. However, their story kernel combined plot-event similarity with character alignment, rather than replacing the full narrative with event chains alone. A similar contrast appears with StoryEmb: [11] report strong performance on both original texts and pseudonymized texts, with P@1 scores of 94.64% and 82.60%, respectively, which indicates that dense task-specific embeddings preserve broader semantic context even when the entity cues are degraded. Our StoryEmb's P@1 also falls from 0.9854 to 0.7736. Overall, these comparisons suggest that explicit structure is most useful when it complements richer narrative information. In this study, zero-shot event and temporo-causal chains improve robustness to entity bias, but lose too much context to match dense narrative embeddings.

5.2 Why Temporo-Causal Enrichment Does Not Improve Retrieval

Contrary to the initial expectation, temporo-causal enrichment did not provide reliable gains over event-only representations. Across embedding architectures, temporal, causal, and joint relation conditions generally failed to improve retrieval and often reduced performance. Hence, assuming that temporo-causality is theoretically relevant, the pipeline may not convert it into discriminative embedding signal, for four possible reasons discussed below.

5.2.1 Relation Labels Reduce Embedding-Space Discriminability. First, relation enrichment reduced discriminability because a small number of highly prevalent relation labels made many summaries look structurally similar. As detailed in Section 4.7.2, the vast majority of inferred links defaulted to minority of categories (such as *Before* and *Cause*) and overwhelmingly connected immediately adjacent events rather than capturing complex discourse. This means that many summaries were represented by similar patterns rather than distinctive long-range narrative structures. The embedding-space analysis in Appendix J.1 supports this interpretation, where average pairwise cosine similarity between summaries mostly increases with each added relation layer. Rather than separating narratives more clearly, the repetitive temporal-causal triples therefore appear to make different stories more similar in embedding space, which degrades the retrieval performance.

5.2.2 Zero-Shot Relation Inference Produces Unstable Outputs. A second explanation concerns the reliability of LLM-based zero-shot relation extraction. Because TMA does not provide gold temporal or causal relation labels, the pipeline could not use supervised

training or balanced in-domain few-shot examples to guide Llama-3's predictions. The relation-inference step therefore required the model to choose among the full temporal, causal and temporo-causal label set in a zero-shot setting. Prior work suggests that such multi-class temporal relation extraction can be difficult for LLMs, and that decomposing the task into simpler binary decisions can improve accuracy [2]. The present results show similar downfalls in the generated relations. As reported in Appendix K.2, the joint temporo-causal condition produced 2,180 invalid relation labels, including out-of-schema labels such as *Statement* and *Cause_Change*. The model also sometimes failed to ground relations in the extracted event list, producing hallucinated event IDs that had to be removed during validation (Appendix K.3). Although removed during post-processing and excluded from the final embeddings, these hallucinated IDs indicate instability in relation inference and might have contributed to relation attrition.

5.2.3 Upstream Event Errors Weaken Relation Enrichment. The third explanation is error propagation from event extraction to relation inference. As [14] note, pipeline-based semantic event extraction can carry earlier errors into later relation stages. In this study, missed or incorrectly typed events may therefore weaken the temporal and causal links inferred by the LLM. The following paragraphs discuss four ways this can happen, moving from what we interpret as the most direct source of upstream weakness to more secondary sources.

Low Performance of Event Type Classification. The manual evaluation in Table 3 shows that the BERT+CRF extractor achieved 72.1 F1 for trigger identification, but only 53.0 F1 for trigger classification on TMA summaries. This indicates that the model often detected that an event was present, but was less reliable in assigning the correct MAVEN event type. This is consistent with MAVEN's own results, where BERT+CRF performance drops from approximately 77.5 F1 on 20% of event types used, to 67.8 F1 when the full 168-type ontology is used [38]. Since the LLM relation-inference step uses these extracted events as input, missed or misclassified events can propagate into the inferred temporal and causal links.

Domain Mismatch of Datasets. MAVEN and TMA are not perfectly aligned, as shown in Figures 2-3. MAVEN contains more real-world event types like *Attack* and *Catastrophe*, while TMA contains more plot-oriented events such as *Know*, *Motion*, and *Arriving*. This mismatch did not prevent event detection altogether, but it likely reduced the reliability of event type classification, thus harming the structural input on which relation enrichment depends.

Reduced Specificity of Event Types. When the Tversky Index is used as a similarity metric instead of cosine similarity of embeddings, we observe that when we represent summaries with event types only, the average pairwise similarity is higher than if we represent them with event triggers (see Appendix I). This suggests that broad event categories increase overlap between unrelated summaries and reduce discriminative power. Because relation inference is performed on these coarse event types, the resulting links are also defined between broad categories rather than between more specific, distinctive events. As a result, the low specificity of the event-type representation is propagated into the relation layer.

Sensitivity to Pseudonymization. We observe that, to a limited extent, event extraction is sensitive to pseudonymization. 81.7% of summaries have at least one trigger-set change after pseudonymization. However, these changes are not substantial. Among all changed summaries, 87.8% of triggers remain the same on average (see Appendix L). Thus, the upstream event extraction may still affect the downstream relation extraction, therefore the retrieval results.

5.2.4 Pseudo-Code-Like Inputs Limit General-Purpose Encoders. The final factor is the fit between the structural linearizations and the embedding models used to encode them. Since Qwen3-0.6B and E5-Mistral-7B perform reasonably on raw narrative text, but degrade on structured conditions, the problem is unlikely to be encoder quality alone. Rather, the linearized inputs combine event IDs, trigger words, MAVEN event types, and relation triples in a pseudo-code-like format (see Appendix G), which may be poorly matched to models trained primarily for natural-language retrieval [36, 41]. It may also help explain why Qwen3-0.6B outperformed E5-Mistral-7B on the structured conditions despite being smaller. Because Qwen3 is also evaluated on code retrieval, unlike the more natural language text-retrieval-focused E5-Mistral, it may be more tolerant of semi-formal structural inputs [36, 41].

5.3 Limitations

5.3.1 Validity and Ground-Truth Constraints. The first limitation concerns construct validity: TMA does not provide gold annotations for narrative events, or temporal and causal relations. As a result, the extracted event-based representations of summaries could not be intrinsically validated at scale. The retrieval experiments therefore assess the incremental downstream contribution of adding temporal and causal relations, rather than directly measuring whether the inferred relations are correct. To provide a partial check on event extraction quality, a manual evaluation was conducted. However, this evaluation was limited to 30 randomly sampled summaries of 5-10 sentences, and was conducted by a single annotator. Therefore, the diagnostic scores should be interpreted cautiously, as no inter-rater agreement could be calculated and statistical robustness is limited.

5.3.2 Computational Overhead and Domain Shift. The second limitation concerns scalability and generalizability. Since we conduct an ablation study with local (large) language models, to manage computational overhead, we had to use cost-effective embedder models like E5-Mistral and Qwen3-0.6B [36, 41]. In addition, because E5-Mistral uses a 4,096 token sliding-window setting, linearized representations longer than 4,096 tokens had to be removed from the pipeline to mitigate data attrition [36]. Furthermore, the framework also depends on the MAVEN ontology [38], which is broad but not tailored to fictional narratives.

5.3.3 Task Decomposition. Finally, relation inference used a fixed zero-shot, multi-class prompt, without systematic optimization of definitions, instructions, or task decomposition. In particular, the lack of task decomposition likely increased the LLM's burden, as it had to select among several temporal and causal labels while producing structured output. As discussed above, this may have contributed to label concentration, invalid labels, and hallucinated event IDs. Following [2], future work could test whether systematic

LLM orchestration and prompt analysis improve relation quality without changing the underlying model.

6 Conclusion

Computational narrative analysis increasingly relies on dense embeddings to identify similarities across stories, yet these state-of-the-art models often exploit surface-level entity bias rather than deep narrative mechanics. This research aimed to answer to what extent explicit narrative modeling—achieved through supervised event extraction combined with LLM-inferred temporo-causal relations—improves narrative similarity retrieval compared to implicit dense embeddings and lexical baselines.

The results show that event-based representations contain meaningful narrative signal, although this signal is limited when events replace the full summary. Under pseudonymization (not otherwise), explicit event-based representations outperform full-text BoW and TF-IDF baselines, which indicates that they are robust under (pseudo-)anonymized conditions (SQ1). Structural enrichment does not reliably improve retrieval over event-only representations: temporal, causal, independent temporal+causal, and joint temporo-causal variants all fail to produce consistent improvements (SQ2). Joint modeling also does not outperform separately inferred temporal and causal structures (SQ3). Although Qwen3-0.6B is stronger than E5-Mistral on linearized structural inputs, relation enrichment remains ineffective across both embedding architectures (SQ4). The best explicit structural model still underperforms StoryEmb and raw-text dense embeddings (SQ5), thus, full-summary embeddings preserve narrative information that is lost when summaries are reduced to event-relation chains. Returning to the main research question, explicit structure is therefore valuable for event selection, but its benefit does not extend to temporo-causal enrichment or to closing the gap with dense narrative models.

These conclusions are qualified by the absence of gold event and relation annotations in TMA, the limited manual validation sample, pipeline cost, the use of a broad MAVEN ontology, and a static zero-shot prompt design. Nevertheless, the robustness of event-based representations under pseudonymization shows that explicit structure remains informative when named-entity shortcuts are removed. The contribution of this thesis is therefore to delimit where explicit narrative structure is useful: event inventories provide a robust and interpretable retrieval signal, while temporal and causal enrichment require more reliable extraction and more discriminative encoding before they can improve performance.

The most promising direction is a hybrid narrative representation that integrates event information with character roles, character arcs, settings, and other discourse features. Such a model could preserve the robustness and interpretability of event inventories while accounting for the narrative dimensions lost in event-relation chains. Future work should support this direction through gold-standard event and relation annotations, narrative-specific event ontologies, domain adaptation, and optimized prompting strategies. In parallel, we see that incorrect matches often share event-type and event-order patterns with the query. This suggests that event sequence similarity deserves further study, especially for cross-lingual or culturally related retellings where plot progression may remain stable despite surface variation.

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Appendix A TMA Test Split Statistics

Figure 4 reports the sentence-length distribution of summaries in the TMA test split. The distribution is skewed to right, summaries contain 17.82 sentences on average, with a median of 12 sentences. Figure 5 shows the distribution of relevance-cluster sizes before and after near-duplicate removal. Deduplication slightly reduces the average number of summaries per story (i.e., distinct work), from 3.70 to 3.31. Table 4 summarizes these corpus-level statistics for the test set of TMA.

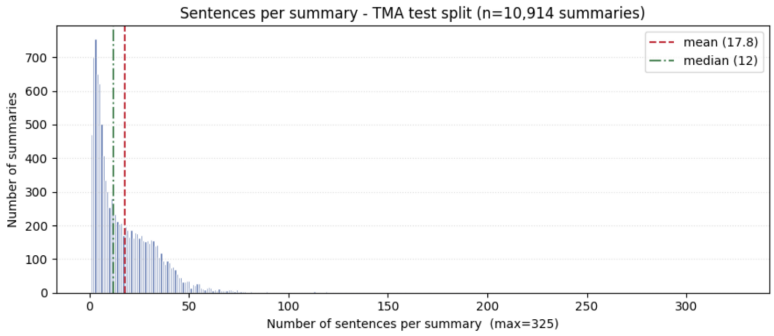


Figure 4: Distribution of the number of sentences per summary in the TMA test split. The dashed red line marks the mean sentence count, while the dash-dotted green line marks the median.

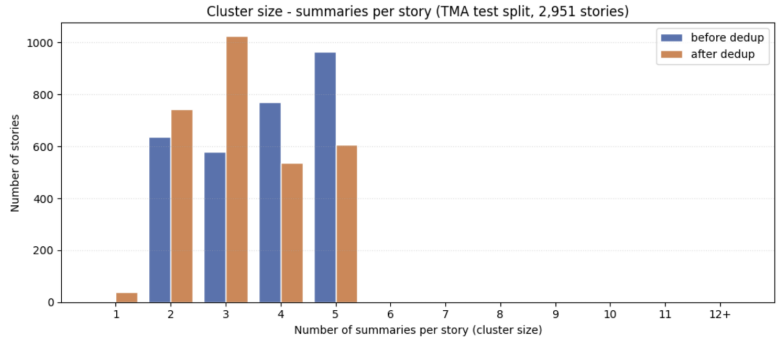


Figure 5: Distribution of summary-cluster sizes per story in the TMA test split before and after near-duplicate removal. Cluster size corresponds to the number of summaries available for the same underlying work.

Table 4: Descriptive statistics for the Tell Me Again! test split before experimental filtering.

Category	Measure	Mean	Median	Min	Max	Std.
Summary structure	Sentences per summary	17.82	12	0	325	18.59
Length	Words per sentence	20.55	18	0	504	11.43
Length	Words per summary	366.06	254	0	6,794	377.51
Cluster size	Raw summaries per story	3.70	4	2	5	1.14
Cluster size	Deduplicated summaries per story	3.31	3	1	5	1.10

Note. The test split contains 2,951 stories and 10,914 summaries. After deduplication, 2,912 stories have at least two summaries and are therefore retrievable, corresponding to 98.7% of stories in the test split.

Appendix B Example Event Extractions from Tell Me Again!

This appendix provides example outputs from the MAVEN-trained BERT+CRF event detector when applied to Tell Me Again! summaries. Event triggers are shown in square brackets, followed by the assigned MAVEN event type. These examples are intended to illustrate the form of the extracted event sequence used as input for subsequent temporal and causal relation inference.

B.1 Example A: Pepe le Moko

s0: Pepe le Moko (Jean Gabin) is a notorious thief who has been **[hiding|Hiding_objects]** in the labyrinthine Casbah for nearly two years.
s1: Despite the police's awareness of his presence, they have been unable to **[capture|Conquering]** him **[due|Causation]** to the community's **[support|Supporting]** and intricate layout.
s2: Pepe's life in the Casbah **[becomes|Becoming]** monotonous, but **[leaving|Departing]** would **[result in|Causation]** his **[arrest|Arrest]**.
s3: Inspector Slimane (Lucas Gridoux) **[forms|Coming_to_be]** an unusual relationship with Pepe and **[vows|Commitment]** to **[arrest|Arrest]** him if he ever tries to **[leave|Departing]**.
s4: One night, Pepe **[meets|Come_together]** the Parisienne, Gaby (Mireille Balin), and **[falls|Motion_directional]** for her.
s5: Meanwhile, Regis, a Casbah resident, **[collaborates|Collaboration]** with the police and **[proposes|Statement]** a **[plan|Arranging]** to **[use|Using]** Pepe's friend Pierrot as bait to lure him out of the Casbah.
s6: Slimane also **[realizes|Know]** Pepe's attraction to Gaby and **[brings|Bringing]** her and her group to the Casbah to **[see|Perception_active]** Pepe.
s7: Tension between Pepe and Regis boils over when Pierrot goes missing, and Pepe **[confronts|Hostile_encounter]** Regis.
s8: Gaby and Pepe **[meet|Come_together]** in the Casbah again and **[spend|Expend_resource]** a night together.
s9: However, when Pierrot **[dies|Death]**, Regis is **[killed|Killing]**, and Gaby **[leaves|Departing]** without saying goodbye.
s10: Pepe **[spirals|Motion]** into a nervous breakdown, but Ines keeps him in the Casbah by **[lying|Placing]** about Gaby's whereabouts.
s11: Pepe later **[finds|Know]** Gaby in the Casbah and they have a passionate **[encounter|Hostile_encounter]**.
s12: They **[agree|Agree_or_refuse_to_act]** to **[see|Perception_active]** each other again the next day.
s13: However, Slimane intervenes and **[tells|Telling]** Gaby's fiance that it is dangerous for her to **[visit|Traveling]** the Casbah, hoping that Pepe will come looking for her if she does not **[show|Presence]** for their rendez-vous.
s14: Despite this **[warning|Warning]**, Gaby **[chooses|Choosing]** to **[ignore|Agree_or_refuse_to_act]** her fiance and **[meet|Come_together]** with Pepe again.
s15: But Slimane **[tells|Telling]** Gaby that Pepe is **[dead|Death]**.
s16: Pepe **[writes|Writing]** a letter to Gaby, which he **[gives|Giving]** to Carlos and **[asks|Request]** him to **[deliver|Sending]** it to her hotel.
s17: When Pepe is later **[informed|Telling]** that Gaby is **[leaving|Departing]** Algiers after being **[told|Telling]** he was **[dead|Death]** by Slimane, he **[leaves|Departing]** the Casbah to find her.
s18: Ines **[tells|Telling]** Slimane about Pepe's **[plan|Arranging]**, **[leading to|Causation]** his **[arrest|Arrest]** at the harbor.
s19: Pepe **[watches|Perception_active]** the ship take Gaby away and **[commits|Killing]** **[suicide|Killing]** with a knife.

B.2 Example B: Night and the City

s0: Henry Fabian, an individual who **[carries out|Action]** a series of shady activities in the London nightclub environment, has as a friend a singer, a good girl who is very much in love with him and whom he unworthily exploits by **[taking|Conquering]** all the money she **[earns|Earnings_and_losses]** from her work.
s1: **[Determined|Deciding]** to wage war on a powerful rival who has a monopoly on wrestling matches and related betting, Fabian **[sets up|Building]** a gym and **[hires|Employment]** an old Greco-Roman wrestling champion.
s2: But things do not go as **[planned|Arranging]**, and Fabian **[finds|Know]** himself up to his neck in trouble.

Appendix C Event-Count Threshold Analysis

Figure 6 shows the distribution of detected MAVEN events per summary in the Tell Me Again! test split. The main experiments use a five-event minimum, which retains 89.7% of summaries. A stricter nine-event threshold would retain 77.1% of summaries. The five-event threshold is therefore selected as the main cutoff because it removes structurally sparse summaries while retaining a substantially larger portion of short summaries for retrieval evaluation.

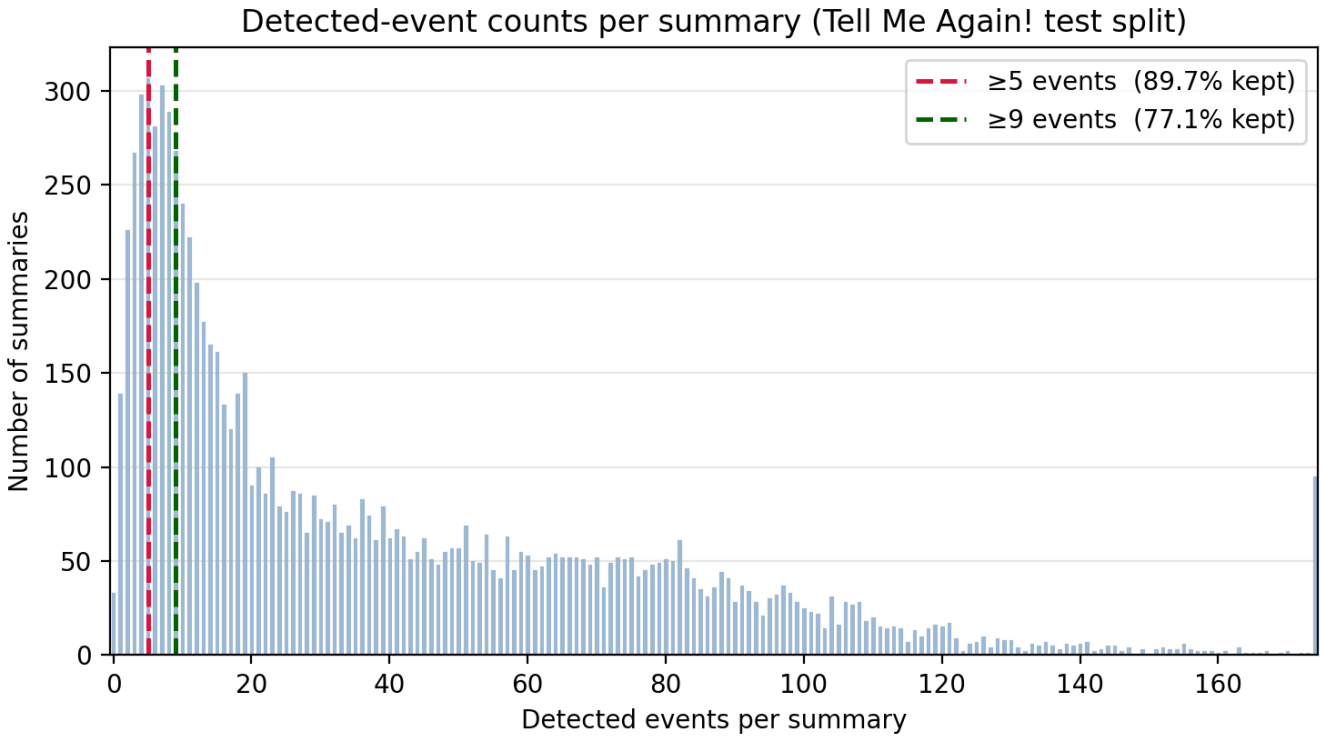


Figure 6: Distribution of detected event counts per summary in the Tell Me Again! test split. The dashed vertical lines show the five-event threshold used in the main experiments and the stricter nine-event threshold considered for comparison.

Appendix D Exemplar JSON Outputs Structural Conditions

This appendix illustrates the structured outputs produced along the representation pipeline, using a single short summary from the *Tell Me Again!* test set. The supervised BERT+CRF model first extracts the events. Then, the LLM is prompted, separately per condition, to infer the relations between these already-extracted events and to return only the relation triples as a strict JSON object. The events themselves are an input to the LLM (inlined in the prompt), not part of its output. Note that the independent temporal+causal condition is an exception: it is not produced by a model call but assembled afterwards.

Input summary. "Commissioner Betti (Maurizio Merli) is transferred to Naples, receiving on his arrival a warm welcome from The Commandante (Barry Sullivan), the city's crime lord. Betti goes on a personal mission against corruption and organized crime, trying to force the syndicate out of town by any means necessary."

Event extraction output. The BERT+CRF detector returns the events below, each with an identifier, its sentence index, the trigger surface form, its MAVEN event type, and the character offsets of the trigger span. The event-only representation consists of exactly these events, with no relations.

```
"events": [
{"event_id": "e1", "sent_id": 0, "trigger": "transferred", "event_type": "Sending",   "start": 39, "end": 50},
{"event_id": "e2", "sent_id": 0, "trigger": "receiving",   "event_type": "Receiving", "start": 62, "end": 71},
{"event_id": "e3", "sent_id": 0, "trigger": "arrival",     "event_type": "Arriving",  "start": 79, "end": 86},
{"event_id": "e4", "sent_id": 1, "trigger": "organized",   "event_type": "Arranging", "start": 56, "end": 65},
{"event_id": "e5", "sent_id": 1, "trigger": "force",       "event_type": "Request",   "start": 83, "end": 88}
]
```

Event + Temporal (e+t). The model is constrained to infer only temporal relations:

```
{"temporal_relations": [
{"source": "e1", "target": "e2", "relation": "BEFORE"}
]}
```

Event + Causal (e+c). The model is constrained to infer only causal relations:

```
{"causal_relations": [
{"source": "e3", "target": "e2", "relation": "ENABLE"}
]}
```

Event + Joint Temporo-Causal (e+t-c). The model is constrained to infer temporo-causal relations (CaTeRs framework):

```
{"joint_relations": [
{"source": "e1", "target": "e2", "relation": "BEFORE"},
{"source": "e2", "target": "e3", "relation": "BEFORE"},
{"source": "e3", "target": "e4", "relation": "ENABLE_BEFORE"},
{"source": "e4", "target": "e5", "relation": "CAUSE_BEFORE"}
]}
```

Event + Temporal + Causal (e+t+c). This condition is not generated by a separate model call. It is assembled post-hoc by merging the outputs of the temporal and causal conditions above. Hence, it contains exactly the relations that the model produced under the temporal and causal prompts.

Appendix E Zero-Shot Prompts Used for Relation Inference

This appendix provides the zero-shot prompt templates used for LLM-based temporal, causal, and joint temporo-causal relation inference. In the pipeline, the supervised BERT+CRF model trained on MAVEN first identifies event triggers and event types. These extracted events are then inlined directly into the narrative summary using the format `[eID|trigger|EVENT_TYPE]`. The LLM is guided by a task-specific role instruction, a fixed relation codebook, and strict JSON formatting constraints. All prompts are wrapped in the Llama chat-template format, using system/user/assistant headers and end-of-turn markers.

E.1 Temporal Relation Prompt

```

<|begin_of_text|><|start_header_id|>system<|end_header_id|>

You are an expert in extracting temporal relations between events. Given a short narrative whose
events have been pre-extracted and inlined as [eID|trigger|EVENT_TYPE] markers,
identify the TEMPORAL relations between these events (if such relation(s) exists).

CODEBOOK: TEMPORAL TAXONOMY
BEFORE:   The source event completely finishes before the target event starts.
OVERLAPS: The source event and target event happen at the same time or intersect.
CONTAINS: The source event completely encapsulates the target event in time.
IDENTITY: The source event and target event refer to the exact same occurrence.

FORMATTING CONSTRAINTS
Output STRICTLY a valid JSON object with this shape:
  {"temporal_relations": [
     {"source": "eID", "target": "eID", "relation": "RELATION_NAME"}
   ]}
Do not include explanations or conversational text.
Refer to events only by the eIDs that appear in the input. Do not invent new IDs.<|eot_id|><|start_header_id|>user<|
    end_header_id|>

INPUT TEXT (events inlined as [eID|trigger|EVENT_TYPE]):
<ANNOTATED_TEXT>

YOUR JSON OUTPUT:<|eot_id|><|start_header_id|>assistant<|end_header_id|>
```

E.2 Causal Relation Prompt

```

<|begin_of_text|><|start_header_id|>system<|end_header_id|>

You are an expert in extracting causal relations between events. Given a short narrative whose
events have been pre-extracted and inlined as [eID|trigger|EVENT_TYPE] markers,
identify the CAUSAL relations between these events (if such relation(s) exists).

CODEBOOK: CAUSAL TAXONOMY
CAUSE:        The source event inevitably causes the target event to happen.
ENABLE:       The source event creates the conditions necessary for the target event.
PREVENT:      The source event stops the target event from happening.
CAUSE_TO_END: The source event causes an ongoing target event to stop.

FORMATTING CONSTRAINTS
Output STRICTLY a valid JSON object with this shape:
  {"causal_relations": [
     {"source": "eID", "target": "eID", "relation": "RELATION_NAME"}
   ]}
Do not include explanations or conversational text.
Refer to events only by the eIDs that appear in the input. Do not invent new IDs.<|eot_id|><|start_header_id|>user<|
    end_header_id|>

INPUT TEXT (events inlined as [eID|trigger|EVENT_TYPE]):
<ANNOTATED_TEXT>

YOUR JSON OUTPUT:<|eot_id|><|start_header_id|>assistant<|end_header_id|>
```

E.3 Joint Temporo-Causal Relation Prompt

```
<|begin_of_text|><|start_header_id|>system<|end_header_id|>

You are an expert in extracting joint temporal-causal relations between events. Given a short narrative whose
events have been pre-extracted and inlined as [eID|trigger|EVENT_TYPE] markers,
identify the JOINT TEMPORO-CAUSAL relations between these events (if such relation(s) exists) using a fused
ontology in which causal relations also encode their temporal implications.

CODEBOOK: JOINT TEMPORO-CAUSAL TAXONOMY
CAUSE_BEFORE:           Source causes target, and source finishes before target starts.
CAUSE_OVERLAPS:         Source causes target, and their time spans intersect.
ENABLE_BEFORE:          Source enables target, and source finishes before target.
ENABLE_OVERLAPS:        Source enables target, and their time spans intersect.
PREVENT_BEFORE:         Source prevents target, and source finishes before target.
PREVENT_OVERLAPS:       Source prevents target, and their time spans intersect.
CAUSE_TO_END_BEFORE:    Source causes target to end, finishing before target ends.
CAUSE_TO_END_OVERLAPS:  Source causes target to end, intersecting in time.
CAUSE_TO_END_DURING:    Source causes target to end entirely during target's span.

If a relation between two events is purely temporal with no causal link, use
one of the following pure-temporal fallback labels (same definitions as the
temporal codebook):
BEFORE:   The source event completely finishes before the target event starts.
OVERLAPS: The source event and target event happen at the same time or intersect.
CONTAINS: The source event completely encapsulates the target event in time.
IDENTITY: The source event and target event refer to the exact same occurrence.

FORMATTING CONSTRAINTS
Output STRICTLY a valid JSON object with this shape:
  {"joint_relations": [
     {"source": "eID", "target": "eID", "relation": "RELATION_NAME"}
   ]}
Do not include explanations or conversational text.
Refer to events only by the eIDs that appear in the input. Do not invent new IDs.<|eot_id|><|start_header_id|>user<|
    end_header_id|>

INPUT TEXT (events inlined as [eID|trigger|EVENT_TYPE]):
<ANNOTATED_TEXT>

YOUR JSON OUTPUT:<|eot_id|><|start_header_id|>assistant<|end_header_id|>
```

Appendix F Exemplar Prompts with Event-Annotated TMA Summaries

This appendix provides exemplar prompt instances used for relation inference after event extraction. Events are inserted directly into the input text using the format `[eID|trigger|EVENT_TYPE]`. The model is instructed to return only a valid JSON object and to use only event identifiers present in the input.

F.1 Temporal Relation Prompt

```
<|begin_of_text|><|start_header_id|>system<|end_header_id|>

You are an expert in extracting temporal relations between events. Given a short narrative whose events have been pre-extracted and inlined as [eID|trigger|EVENT_TYPE] markers, identify the TEMPORAL relations between these events (if such relation(s) exists).

CODEBOOK: TEMPORAL TAXONOMY
BEFORE:   The source event completely finishes before the target event starts.
OVERLAPS: The source event and target event happen at the same time or intersect.
CONTAINS: The source event completely encapsulates the target event in time.
IDENTITY: The source event and target event refer to the exact same occurrence.

FORMATTING CONSTRAINTS
Output STRICTLY a valid JSON object with this shape:
  {"temporal_relations": [
     {"source": "eID", "target": "eID", "relation": "RELATION_NAME"}
   ]}
Do not include explanations or conversational text.
Refer to events only by the eIDs that appear in the input. Do not invent new IDs.<|eot_id|><|start_header_id|>user<|end_header_id|>

INPUT TEXT (events inlined as [eID|trigger|EVENT_TYPE]):
A woman [e1|suspected|Suspicion] of being the material author of a [e2|murder|Killing] undergoes plastic surgery and [e3|changes|Change] her face. When you [e4|change|Change] your face, try to [e5|change|Change] your life.

YOUR JSON OUTPUT:<|eot_id|><|start_header_id|>assistant<|end_header_id|>
```

F.2 Causal Relation Prompt

```
<|begin_of_text|><|start_header_id|>system<|end_header_id|>

You are an expert in extracting causal relations between events. Given a short narrative whose events have been pre-extracted and inlined as [eID|trigger|EVENT_TYPE] markers, identify the CAUSAL relations between these events (if such relation(s) exists).

CODEBOOK: CAUSAL TAXONOMY
CAUSE:        The source event inevitably causes the target event to happen.
ENABLE:       The source event creates the conditions necessary for the target event.
PREVENT:      The source event stops the target event from happening.
CAUSE_TO_END: The source event causes an ongoing target event to stop.

FORMATTING CONSTRAINTS
Output STRICTLY a valid JSON object with this shape:
  {"causal_relations": [
     {"source": "eID", "target": "eID", "relation": "RELATION_NAME"}
   ]}
Do not include explanations or conversational text.
Refer to events only by the eIDs that appear in the input. Do not invent new IDs.<|eot_id|><|start_header_id|>user<|end_header_id|>

INPUT TEXT (events inlined as [eID|trigger|EVENT_TYPE]):
A woman [e1|suspected|Suspicion] of being the material author of a [e2|murder|Killing] undergoes plastic surgery and [e3|changes|Change] her face. When you [e4|change|Change] your face, try to [e5|change|Change] your life.

YOUR JSON OUTPUT:<|eot_id|><|start_header_id|>assistant<|end_header_id|>
```

F.3 Joint Temporo-Causal Relation Prompt

```
<|begin_of_text|><|start_header_id|>system<|end_header_id|>

```

```
You are an expert in extracting joint temporal-causal relations between events. Given a short narrative whose events have been pre-extracted and inlined as [eID|trigger|EVENT_TYPE] markers, identify the JOINT TEMPORO-CAUSAL relations between these events (if such relation(s) exists) using a fused ontology in which causal relations also encode their temporal implications.

CODEBOOK: JOINT TEMPORO-CAUSAL TAXONOMY
CAUSE_BEFORE:           Source causes target, and source finishes before target starts.
CAUSE_OVERLAPS:         Source causes target, and their time spans intersect.
ENABLE_BEFORE:          Source enables target, and source finishes before target.
ENABLE_OVERLAPS:        Source enables target, and their time spans intersect.
PREVENT_BEFORE:         Source prevents target, and source finishes before target.
PREVENT_OVERLAPS:       Source prevents target, and their time spans intersect.
CAUSE_TO_END_BEFORE:    Source causes target to end, finishing before target ends.
CAUSE_TO_END_OVERLAPS:  Source causes target to end, intersecting in time.
CAUSE_TO_END_DURING:    Source causes target to end entirely during target's span.

If a relation between two events is purely temporal with no causal link, use one of the following pure-temporal fallback labels:
BEFORE:   The source event completely finishes before the target event starts.
OVERLAPS: The source event and target event happen at the same time or intersect.
CONTAINS: The source event completely encapsulates the target event in time.
IDENTITY: The source event and target event refer to the exact same occurrence.

FORMATTING CONSTRAINTS
Output STRICTLY a valid JSON object with this shape:
  {"joint_relations": [
     {"source": "eID", "target": "eID", "relation": "RELATION_NAME"}
   ]}
Do not include explanations or conversational text.
Refer to events only by the eIDs that appear in the input. Do not invent new IDs.<|eot_id|><|start_header_id|>user<|end_header_id|>

INPUT TEXT (events inlined as [eID|trigger|EVENT_TYPE]):
A woman [e1|suspected|Suspicion] of being the material author of a [e2|murder|Killing] undergoes plastic surgery and [e3|changes|Change] her face. When you [e4|change|Change] your face, try to [e5|change|Change] your life.

YOUR JSON OUTPUT:<|eot_id|><|start_header_id|>assistant<|end_header_id|>
```

Appendix G Example Linearized Representations

This appendix shows the six structural representation conditions used before embedding, with one exemplar summary. The `raw text` condition keeps the original summary.

G.1 Raw Text

```

Commissioner Betti (Maurizio Merli) is transferred to Naples, receiving on his arrival a warm welcome from The
    Commandante (Barry Sullivan), the city's crime lord. Betti goes on a personal mission against corruption and
    organized crime, trying to force the syndicate out of town by any means necessary.
```

G.2 Events Only Linearization

```
EVENTS:
(e1|transferred|Sending) (e2|receiving|Receiving) (e3|arrival|Arriving) (e4|organized|Arranging) (e5|force|Request)
```

G.3 Temporal Linearization

```
EVENTS:
(e1|transferred|Sending) (e2|receiving|Receiving) (e3|arrival|Arriving) (e4|organized|Arranging) (e5|force|Request)

TEMPORAL:
(e1:transferred|Sending, BEFORE, e2:receiving|Receiving)
```

G.4 Causal Linearization

```
EVENTS:
(e1|transferred|Sending) (e2|receiving|Receiving) (e3|arrival|Arriving) (e4|organized|Arranging) (e5|force|Request)

CAUSAL:
(e3:arrival|Arriving, ENABLE, e2:receiving|Receiving)
```

G.5 Joint Temporo-Causal Linearization

```
EVENTS:
(e1|transferred|Sending) (e2|receiving|Receiving) (e3|arrival|Arriving) (e4|organized|Arranging) (e5|force|Request)

TEMPOROCAUSAL:
(e1:transferred|Sending, BEFORE, e2:receiving|Receiving)
(e2:receiving|Receiving, BEFORE, e3:arrival|Arriving)
(e3:arrival|Arriving, ENABLE_BEFORE, e4:organized|Arranging)
(e4:organized|Arranging, CAUSE_BEFORE, e5:force|Request)
```

G.6 Independent Temporal+Causal Linearization

```
EVENTS:
(e1|transferred|Sending) (e2|receiving|Receiving) (e3|arrival|Arriving) (e4|organized|Arranging) (e5|force|Request)

TEMPORAL:
(e1:transferred|Sending, BEFORE, e2:receiving|Receiving)

CAUSAL:
(e3:arrival|Arriving, ENABLE, e2:receiving|Receiving)
```

Appendix H Structural Confusion Error Analyses

H.1 Structural confusion across encoder–condition pairs

Table 5 reports the structural confusion diagnostics across all encoder-condition pairs. For each incorrect top-1 retrieval, we compare the wrongly retrieved summary with a random non-matching summary using event-type and trigger overlap measures. Inventory overlap is computed as the Dice overlap between the two sets of event types or triggers. Order overlap is computed using the longest common subsequence (LCS) between the two event-type or trigger sequences, normalized by the combined sequence lengths. For each measure, we test whether wrongly retrieved pairs exhibit higher overlap than random non-matching pairs using a one-sided Mann–Whitney U test, then we report the error-pair mean, the random-pair mean, their difference, and the resulting p-value.

Table 5: Structural similarity of incorrect top-1 retrievals compared with random non-matching summaries across all encoder–condition pairs.

Encoder	Condition	Wrong@1	Measure	Error mean	Random mean	Δ	p
Qwen3-0.6B	Event-only	3382/4315 (78.4%)	Event-type inventory overlap	0.4319	0.2295	+0.2024	< .001
			Event-type order overlap	0.2010	0.1116	+0.0894	< .001
			Trigger inventory overlap	0.0980	0.0364	+0.0616	< .001
			Trigger order overlap	0.0644	0.0259	+0.0385	< .001
Qwen3-0.6B	Temporal	3427/4315 (79.4%)	Event-type inventory overlap	0.4319	0.2321	+0.1999	< .001
			Event-type order overlap	0.2063	0.1137	+0.0927	< .001
			Trigger inventory overlap	0.0979	0.0364	+0.0615	< .001
			Trigger order overlap	0.0652	0.0263	+0.0389	< .001
Qwen3-0.6B	Causal	3459/4315 (80.2%)	Event-type inventory overlap	0.4244	0.2318	+0.1926	< .001
			Event-type order overlap	0.2006	0.1127	+0.0879	< .001
			Trigger inventory overlap	0.0956	0.0362	+0.0594	< .001
			Trigger order overlap	0.0640	0.0256	+0.0384	< .001
Qwen3-0.6B	Temporal+causal independent	3421/4315 (79.3%)	Event-type inventory overlap	0.4304	0.2305	+0.1999	< .001
			Event-type order overlap	0.2060	0.1120	+0.0940	< .001
			Trigger inventory overlap	0.0975	0.0369	+0.0606	< .001
			Trigger order overlap	0.0660	0.0263	+0.0397	< .001
Qwen3-0.6B	Joint temporo-causal	3437/4315 (79.7%)	Event-type inventory overlap	0.4270	0.2309	+0.1961	< .001
			Event-type order overlap	0.2044	0.1126	+0.0917	< .001
			Trigger inventory overlap	0.0973	0.0379	+0.0594	< .001
			Trigger order overlap	0.0656	0.0270	+0.0387	< .001
E5-Mistral	Event-only	3608/4315 (83.6%)	Event-type inventory overlap	0.3956	0.2356	+0.1600	< .001
			Event-type order overlap	0.2112	0.1146	+0.0967	< .001
			Trigger inventory overlap	0.0976	0.0388	+0.0589	< .001
			Trigger order overlap	0.0715	0.0272	+0.0443	< .001
E5-Mistral	Temporal	4005/4315 (92.8%)	Event-type inventory overlap	0.3869	0.2349	+0.1520	< .001
			Event-type order overlap	0.1992	0.1129	+0.0863	< .001
			Trigger inventory overlap	0.0868	0.0378	+0.0490	< .001
			Trigger order overlap	0.0615	0.0267	+0.0348	< .001
E5-Mistral	Causal	4021/4315 (93.2%)	Event-type inventory overlap	0.3907	0.2378	+0.1528	< .001
			Event-type order overlap	0.2005	0.1140	+0.0865	< .001
			Trigger inventory overlap	0.0876	0.0378	+0.0498	< .001
			Trigger order overlap	0.0613	0.0267	+0.0346	< .001
E5-Mistral	Temporal+causal independent	4157/4315 (96.3%)	Event-type inventory overlap	0.3686	0.2378	+0.1308	< .001
			Event-type order overlap	0.1897	0.1135	+0.0762	< .001
			Trigger inventory overlap	0.0760	0.0389	+0.0371	< .001
			Trigger order overlap	0.0533	0.0275	+0.0258	< .001
E5-Mistral	Joint temporo-causal	3994/4315 (92.6%)	Event-type inventory overlap	0.3916	0.2390	+0.1526	< .001
			Event-type order overlap	0.2025	0.1145	+0.0880	< .001

Continued on next page

Encoder	Condition	Wrong@1	Measure	Error mean	Random mean	Δ	p
			Trigger inventory overlap	0.0896	0.0383	+0.0513	< .001
			Trigger order overlap	0.0628	0.0270	+0.0358	< .001

H.2 Exemplar Structural Confusion Cases

This appendix reports two exemplar top-1 retrieval errors from the Qwen3-0.6B event-only condition. The cases are selected by highest event-type order overlap among incorrect top-1 retrievals. They illustrate cases where the retrieved summary is not from the same underlying work as the query, but nevertheless shares a similar high-level event sequence skeleton.

Table 6: Case 1 - Shared movement and discovery skeleton

Query summary	Wrongly retrieved summary
A young man, Paul Harrison, the son of a wealthy British businessman living in Paris, meets a beautiful young orphan, Michelle Latour. The two teenagers leave Paris for the Camargue. Michelle became pregnant and gave birth to a baby girl. The young couple and the child lead a family life until the police find them.	An army veteran, the sole survivor of a decimated American patrol, meets a young South Korean, Short Round, as well as others left behind by the war. He leads them to an unoccupied Buddhist temple, which they turn into an observation camp. But when they discover that they are in the immediate vicinity of a North Korean communist camp, the troop prepares for the possibility of a fight.
Event types: Come together, Departing, Becoming, Giving, Know.	**Event types:** Destroying, Come together, Departing, Becoming, Know, GetReady.
Event triggers: meets, leave, became, gave, find.	**Event triggers:** decimated, meets, left, turn, discover, prepares.

Note. Cosine similarity = 0.797; event-type order overlap = 0.727; event-type inventory overlap = 0.727; trigger inventory overlap = 0.182. Shared event-type skeleton: Come together, Departing, Becoming, Know.

Table 7: Case 2 - Shared transfer-arrival-warning skeleton

Query summary	Wrongly retrieved summary
Doctor Gerda Maurer is found guilty of negligence and sent to prison. On her release she cannot return to her career in medicine and has to take on other jobs such as a nightclub singer. She is giving a second chance by a respected surgeon and works as his assistant, but her new position is threatened by the return of her former lover Georg.	Commissioner Betti is being transferred to Naples on orders from his hierarchy. As soon as he arrives on the scene, the local mafia gives him a threatening welcome in the form of a warning. The latter is placed under the di O Generale cup. At his instigation, the organization becomes the scene of a fratricidal reckoning.
Event types: Know, Sending, Releasing, Arriving, Giving, Warning.	**Event types:** Sending, Arriving, Giving, Warning, Placing, Becoming.
Event triggers: found, sent, release, return, giving, threatened.	**Event triggers:** transferred, arrives, gives, warning, placed, becomes.

Note. Cosine similarity = 0.811; event-type order overlap = 0.667; event-type inventory overlap = 0.667; trigger inventory overlap = 0.000. Shared event-type skeleton: Sending, Arriving, Giving, Warning.

Appendix I Event-Overlap Diagnostic Baselines

We compute set-overlap baselines over extracted event information. Each summary is represented as an unordered set of either MAVEN event types or event-triggers, and pairwise similarity is computed using the Dice coefficient. These baselines test whether retrieval signal is present in extracted events without dense embeddings, TF-IDF weighting, or relation structure. Together, these baselines separate three sources of retrieval signal: overlap in the original summary text, overlap in extracted trigger words, and overlap in broader MAVEN event categories.

Table 8: Diagnostic retrieval performance of set-based event representations using the Tversky index on the matched evaluation set.

Story representation	Vocab. size	Mean items	SD items	Min	Max	Mean Sim.	SD Sim.	P@1	Hits@10	R-Prec.	MAP	NDCG
Non-Pseudonymized												
Event types (MAVEN categories)	153	23.3	13.2	3	60	0.238	0.135	0.1495	0.2401	0.1076	0.1229	0.2581
Events (trigger words)	6174	32.7	22.4	4	107	0.038	0.042	0.2350	0.3506	0.1710	0.1951	0.3324
Pseudonymized												
Event types (MAVEN categories)	152	22.4	12.8	3	59	0.232	0.134	0.1423	0.2308	0.1031	0.1182	0.2521
Events (trigger words)	6031	31.1	21.4	4	99	0.037	0.042	0.2176	0.3384	0.1597	0.1825	0.3193

Note. Results are computed on the matched intersection of the non-pseudonymized and pseudonymized evaluation sets, so that both conditions are evaluated on the same 4,315 summaries from 1,658 stories. The vocabulary sizes therefore refer only to event types or trigger words observed in this matched intersection, yet they can differ between two arms (see Appendix L). Similarity is computed using the symmetric Tversky index with $\alpha = \beta = 0.5$, equivalent to the Dice coefficient. Candidates are ranked by this score, with self-matches excluded. 'Mean items' and 'SD items' refer to the mean and standard deviation of the number of distinct event types or trigger words per summary. 'Mean sim.' and 'SD sim.' refer to the mean and standard deviation of off-diagonal pairwise similarity. Higher mean similarity indicates lower discriminatory power because unrelated summaries are more likely to overlap.

Appendix J Relation-Mismatch Diagnostic Tables

This appendix reports the diagnostic tables supporting the analysis of structural mismatch from relation enrichment. The diagnostics examine whether temporal and causal relation layers add discriminative narrative structure or instead make representations similar to each other by adding imbalanced relation patterns.

J.1 Embedding-Space Collapse

Table 9: Pairwise cosine similarity by encoder and representation condition.

Encoder	Condition	Mean cosine	SD	Min	Max
Qwen3-0.6B	Event-only	0.6183	0.0941	0.2234	0.9812
Qwen3-0.6B	Temporal	0.6459	0.0897	0.2635	0.9822
Qwen3-0.6B	Causal	0.6502	0.0889	0.2623	0.9839
Qwen3-0.6B	Temporal + causal	0.6588	0.0844	0.2753	0.9787
Qwen3-0.6B	Joint temporo-causal	0.6695	0.0801	0.2934	0.9815
E5-Mistral	Event-only	0.8890	0.0279	0.7095	0.9958
E5-Mistral	Temporal	0.9070	0.0378	0.5220	0.9944
E5-Mistral	Causal	0.9140	0.0281	0.6674	0.9956
E5-Mistral	Temporal + causal	0.9237	0.0416	0.5776	0.9943
E5-Mistral	Joint temporo-causal	0.9169	0.0315	0.5250	0.9943

Note. Pairwise cosine statistics are computed over off-diagonal pairwise similarities between all summary embeddings within each condition. Higher mean cosine values indicate a more closeness in embedding space, where summaries are less clearly separated from one another. 'SD' refers to standard deviation.

J.2 Relation-Label Diversity

Table 10 shows that the inferred relation labels are densely repeated. Most temporal links are labeled *Before*, most causal links are labeled *Cause*, and most joint temporo-causal links are labeled *Cause_Before*. This limited label diversity suggests that relation enrichment often adds repeated structural patterns rather than strongly discriminative information.

Table 10: Distribution of relation labels by structural condition.

Condition	Label	Count	Share
Temporal	BEFORE	93605	84.8%
Temporal	IDENTITY	10534	9.5%
Temporal	CONTAINS	3728	3.4%
Temporal	OVERLAPS	2575	2.3%
Causal	CAUSE	54186	91.1%
Causal	ENABLE	2878	4.8%
Causal	PREVENT	1249	2.1%
Causal	CAUSE_TO_END	1149	1.9%
Temporal + causal	BEFORE	93605	55.1%
Temporal + causal	CAUSE	54186	31.9%
Temporal + causal	IDENTITY	10534	6.2%
Temporal + causal	CONTAINS	3728	2.2%
Temporal + causal	ENABLE	2878	1.7%
Temporal + causal	OVERLAPS	2575	1.5%
Temporal + causal	PREVENT	1249	0.7%
Temporal + causal	CAUSE_TO_END	1149	0.7%
Joint temporo-causal	CAUSE_BEFORE	59543	72.9%
Joint temporo-causal	CAUSE_TO_END_BEFORE	7564	9.3%
Joint temporo-causal	ENABLE_BEFORE	4552	5.6%
Joint temporo-causal	CAUSE_OVERLAPS	4534	5.5%
Joint temporo-causal	BEFORE	1703	2.1%
Joint temporo-causal	PREVENT_BEFORE	1646	2.0%
Joint temporo-causal	CAUSE_TO_END_DURING	792	1.0%
Joint temporo-causal	CONTAINS	374	0.5%
Joint temporo-causal	ENABLE_OVERLAPS	337	0.4%
Joint temporo-causal	IDENTITY	314	0.4%
Joint temporo-causal	CAUSE_TO_END_OVERLAPS	263	0.3%
Joint temporo-causal	PREVENT_OVERLAPS	54	0.1%
Joint temporo-causal	OVERLAPS	18	0.0%

Note. Shares are computed within each relation condition. The temporal + causal condition combines independently inferred temporal and causal links.

J.3 Link-Pattern Predictability

Table 11 summarizes where inferred relation links appear in the event sequence. *Relations* gives the total number of inferred links for each condition. *Adjacent* is the percentage of links between consecutive events, and *Forward* is the percentage of links where the source event occurs before the target event. High values in both columns indicate that most inferred relations follow the local left-to-right event order.

Table 11: Adjacency and directionality of inferred relation links.

Condition	Relations	Adjacent	Forward
Temporal	110442	98.8%	96.0%
Causal	59462	91.9%	96.1%
Temporal + causal	169904	96.4%	96.0%
Joint temporo-causal	81694	95.5%	94.6%

Note. Adjacent links connect events whose indices differ by one. Forward links connect a source event to a later target event in the extracted event sequence.

J.4 Top-1 Retrieval Casualty Analysis

Table 12: Top-1 retrieval changes after adding relation structure to event-only representations.

Encoder	Condition	Broken	Fixed	Net
Qwen3-0.6B	Temporal	158	113	-45
Qwen3-0.6B	Causal	222	145	-77
Qwen3-0.6B	Temporal + causal	198	159	-39
Qwen3-0.6B	Joint temporo-causal	188	133	-55
E5-Mistral	Temporal	450	53	-397
E5-Mistral	Causal	479	66	-413
E5-Mistral	Temporal + causal	572	23	-549
E5-Mistral	Joint temporo-causal	433	47	-386

Note. The reference condition is the event-only representation for the same encoder. Broken counts queries that event-only answered correctly but the relation-enriched condition retrieves incorrectly. Fixed counts queries that event-only answered incorrectly but the relation-enriched condition retrieves correctly. Net is Fixed minus Broken; negative values indicate that the relation layer breaks more top-1 retrievals than it repairs.

Appendix K Data Attrition

This appendix reports the diagnostic tables used to quantify corpus attrition and annotation noise across the event-relation pipeline.

K.1 Corpus Attrition Across Pipeline Stages

Table 13: Summary-level attrition across the experimental pipeline.

Stage	Cause	Dropped	Summaries out
Pipeline S2 subsetting	On official test split: genre, translation-deduplication, length, BERT-subword, and at least two summaries per work filters	–	9385
Pipeline S3.5 pre-annotation	Fewer than 5 detected events	968 (10.3%)	8417
Pipeline S3.5 pre-annotation	Cluster floor: work left with fewer than 2 summaries	195 (2.1%)	8222
Pipeline S4.5 complete-case	Llama relations missing in at least one condition (parse error or context overflow; see Thesis Section 3.2)	1666 (20.3%)	6556
Pipeline S4.5 complete-case	Cluster floor: work left with fewer than 2 summaries	367 (4.5%)	6189
Pipeline S5.5 linearization token limit	Linearized string over 4096 tokens in any condition	391 (6.3%)	5798
Pipeline S5.5 linearization token limit	Cluster floor: work left with fewer than 2 summaries	132 (2.1%)	5666
Evaluation matched intersection	Keep only summaries surviving in both the non-pseudonymized and pseudonymized arms	1270 (22.4%)	4396
Evaluation Notebook matched intersection	Cluster floor: work left with fewer than 2 summaries	81 (1.4%)	4315

Note. The Stage column refers to the corresponding sections of the `Pipeline` and `Evaluation` notebooks in the GitHub repository, with section denoted as 'S'. The table reports all stages starting from the non-pseudonymized arm, and the same filtering decisions are then applied to the pseudonymized arm. Counts up to the linearization limit are computed on the non-pseudonymized version of the dataset. Percentages are computed relative to the input size of the corresponding stage. The final matched-intersection row restricts evaluation to summaries present in both the non-pseudonymized and pseudonymized arms.

K.2 Invalid Relation Labels

Table 14: Top-8 invalid relation labels and malformed relation entries dropped during validation.

Condition	Dropped as	Count
Temporal	Invalid label AFTER	1760
Temporal	Invalid label CAUSATION	68
Temporal	Invalid label CAUSES	26
Temporal	Invalid label MEETS	4
Temporal	Invalid label REFUSES	3
Temporal	Invalid label MEET	3
Temporal	Invalid label AFFECTS	3
Temporal	Invalid label EXCHANGE	2
Temporal	Other invalid labels (7 types)	8
Temporal	Malformed entry / bad eID format	1
Temporal	TOTAL dropped	1878 (0.78% of emitted)
Causal	Invalid label CAUSATION	31
Causal	Invalid label HINDERING	22
Causal	Invalid label HINDER	8
Causal	Invalid label CREATE	7
Causal	Invalid label CAUSE_TO_MAKE_PROGRESS	7
Causal	Invalid label CAUSE_CHANGE_OF_POSITION_ON_A_SCALE	6
Causal	Invalid label CAUSE_TO_AMALGAMATE	5
Causal	Invalid label STATEMENT	4
Causal	Other invalid labels (21 types)	29
Causal	Malformed entry / bad eID format	51
Causal	TOTAL dropped	170 (0.14% of emitted)
Temporal-causal joint	Invalid label STATEMENT	289
Temporal-causal joint	Invalid label CAUSE_CHANGE_OF_POSITION_ON_A_SCALE	149
Temporal-causal joint	Invalid label CAUSE_CHANGE	70
Temporal-causal joint	Invalid label NAME_CONFERRAL	66
Temporal-causal joint	Invalid label CAUSE	61
Temporal-causal joint	Invalid label CAUSE_TO_AMALGAMATE	60
Temporal-causal joint	Invalid label CAUSE_TO_BE_INCLUDED	58
Temporal-causal joint	Invalid label CAUSE_TO_MAKE_PROGRESS	44
Temporal-causal joint	Other invalid labels (680 types)	1383
Temporal-causal joint	Malformed entry / bad eID format	1
Temporal-causal joint	TOTAL dropped	2181 (1.43% of emitted)

Note. Invalid labels are relation labels emitted by the LLM that do not belong to the predefined relation ontology for the corresponding condition. The "Other invalid labels" row collapses all remaining out-of-codebook labels beyond the eight most frequent. Malformed entries are structurally invalid relation records, including bad event-ID formatting. The TOTAL dropped row equals the sum of all rows above it (invalid labels + collapsed + malformed), and its percentage is relative to all emitted relations.

K.3 Hallucinated Event Identifiers

Table 15: Hallucinated event identifiers by relation condition.

Condition	Hallucinated	Kept	Rate	Exactly e(N+1)
Temporal	677	110442	0.61%	96.8%
Causal	247	59462	0.41%	100.0%
Temporal-causal joint	518	81694	0.63%	99.8%
Temporal-causal independent	924	169904	0.54%	97.6%

Note. *Hallucinated* counts relation links that refer to event IDs not present in the extracted event list; these relations are dropped before linearization. *Kept* counts valid relation links retained after this filtering, and *Rate* is the percentage of hallucinated links among all emitted links. *Exactly e(N+1)* reports the share of hallucinated IDs that exactly match the next unused event ID after the final valid extracted event.

Appendix L Pseudonymization Effects on Event-Trigger Extraction

Here, we demonstrate why event-trigger BoW and TF-IDF are not perfectly identical across the non-pseudonymized and pseudonymized settings. Since we run the full pipeline separately for both dataset variants, event extraction is applied to each of them separately. Then, we observe that replacing named entities with generic placeholders can slightly change the context available to the MAVEN-trained BERT+CRF extractor and therefore alter its predicted triggers. After pseudonymization, 81.7% of summaries have at least one trigger change. However, the changes are usually small: across all summaries, the trigger sets are 90.0% identical on average, and even among the summaries with at least one changed trigger, they are still 87.8% identical on average.

Table 16: Event-trigger drift after pseudonymization on the matched evaluation set.

Measure	Value
Matched summaries	4,315
Summaries with changed trigger set	3,527 (81.7%)
Mean trigger-set Jaccard overlap, all summaries	90.0%
Mean trigger-set Jaccard overlap, changed summaries only	87.8%

Note. A changed trigger set means that at least one extracted trigger differs between the non-pseudonymized and pseudonymized versions of the same summary. Jaccard overlap measures how similar the two trigger sets are; 100% means identical trigger sets.

Table 17: Example event-trigger changes after pseudonymization.

Example	Non-pseudonymized	Pseudonymized
The Dangerous Lives of Altar Boys	**Non-pseudonymized-only triggers:** read, study, transition	**Pseudonymized-only triggers:** –
(741378 / es)	The plot is **[set|Placing]** in the 1970s, and depicts the lives of two teenagers, Tim Sullivan (Kieran Culkin) and Francis Doyle (Emile Hirsch), who in the comics they **[read|Perception_active]**, make and **[find|Know]** an **[escape|Escaping]** from the strict discipline they **[suffer|Bodily_harm]** in the school of Catholic nuns where they **[study|Education_teaching]**. Particularly, the biggest pressure comes from Sister Assumpta (Jodie Foster), who **[argues|Quarreling]** that she is concerned for their souls, but only annoys the boys with continuous **[scolding|Judgment_communication]**. Tim and Francis, as a secret **[revenge|Revenge]**, draw her as Peg Leg, the main villain in their comic strips, which also reflect much of her helplessness and despair over life at school, and the difficult **[transition|Change]** from childhood to adolescence.	The plot is **[set|Placing]** in the 1970s, and depicts the lives of two teenagers, Byron and Alisha, who in the comics Anton read, make and **[find|Know]** an **[escape|Escaping]** from the strict discipline Anton **[suffer|Bodily_harm]** in the school of Entity O nuns where Anton study. Particularly, the biggest pressure comes from Sister Wiley (Phyllis), who **[argues|Quarreling]** that she is concerned for Anton's souls, but only annoys Anton with continuous **[scolding|Judgment_communication]**. Anton, as a secret **[revenge|Revenge]**, draw her as Cierra, the main villain in Anton's comic strips, which also reflect much of her helplessness and despair over life at school, and the difficult transition from childhood to adolescence.
Boss Level	**Non-pseudonymized-only triggers:** save	**Pseudonymized-only triggers:** death
(54894945 / fr)	Trapped in a time loop that constantly repeats the day of his murder, former Special Forces agent Roy Pulver (Frank Grillo) **[discovers|Know]** clues about a secret government project that could unravel the mystery of his untimely death. In a **[race|Competition]** against time, Pulver must track down Colonel Clive Ventor (Mel Gibson), the powerful head of the government program, while outmaneuvering skilled and ruthless assassins **[determined|Deciding]** to divert him from the truth in order to **[break out|Process_start]** of the loop, **[save|Rescuing]** his wife (Naomi Watts) and live again for tomorrow.	Trapped in a time loop that constantly repeats the day of his murder murder, Sharon **[discovers|Know]** clues about a secret government project that could unravel the mystery of his untimely **[death|Death]** untimely **[death|Death]**. In a **[race|Competition]** against time, Sharon must track down Colonel James (Oliver), the powerful head of the government program, while outmaneuvering skilled and ruthless assassins **[determined|Deciding]** to divert Sharon from the truth in order to **[break out|Process_start]** of the loop, save Sharon's wife (Taylor) and live again for tomorrow.

Note. Bracketed spans indicate extracted triggers and MAVEN event types. The examples are sampled from summaries with at most three sentences whose trigger set differs after pseudonymization. Each example is given with its Wikidata ID of the work and a summary ID, which is the language code, since each work in TMA has at most one summary per language and every summary originates from a different-language Wikipedia edition (here translated into English). The language code therefore uniquely identifies which of the work's summaries is shown.

Appendix M Use of Generative AI

Generative AI was used as a support tool during the preparation of this thesis in several ways. First, LLama model was used as part of the study design to support the extraction of temporal and causal relationships between events. Second, Claude was used to assist with some of the coding-related tasks, including debugging, identifying potential coding issues, and examining the problems within the source code of external models and APIs used in this study, during the reproducibility stages. Third, ChatGPT was used for writing-related support, such as brainstorming possible structures for the thesis, however, never for the final decisions. It was also used for revising LaTeX tables and captions, checking readability, and improving the clarity of textual and visual presentations.